2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report is generated from completed solver output directories. Tables, figures, and quantitative qualifications are traceable through the run and figure manifests; no force, residual, or field values are entered by hand.

1 Introduction and cases

The benchmark covers inviscid and laminar NACA0012 cases, a steady cylinder case at Reynolds 20, and a physical-time cylinder calculation at Reynolds 200. The run-status table in the generated results identifies the actual submitted directory, MPI rank count, wall time, and completion qualification for each.

2 Governing equations and nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the two-dimensional compressible Navier–Stokes equations are

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}^i + \partial_y \mathbf{G}^i = \partial_x \mathbf{F}^v + \partial_y \mathbf{G}^v.$$

The inviscid fluxes are $\mathbf{F}^i = [\rho u, \rho u^2{+}p, \rho uv, u(\rho E{+}p)]^T$ and $\mathbf{G}^i = [\rho v, \rho uv, \rho v^2{+}p, v(\rho E{+}p)]^T$. The closure is $p = (\gamma-1)\rho e$, $E = e+(u^2+v^2)/2$, and $a = \sqrt{\gamma p/\rho}$. Coefficients use the case-file reference dynamic pressure $q_\infty = \rho_\infty U_\infty^2/2$, length, and area. Laminar viscosity is $\mu = \rho_\infty U_\infty L_{Re}/Re$ and $k = \mu c_p/Pr$. The implemented Newtonian and Fourier terms are

$$\boldsymbol{\tau} = \mu\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^T - \frac{2}{3}(\nabla\cdot\boldsymbol{u})\mathbf{I}\right), \qquad \boldsymbol{q} = -k\nabla T,$$

so the viscous face flux in the conservative-equation convention is

$$\mathbf{F}^v\cdot\boldsymbol{n} = [0, (\boldsymbol{\tau}\boldsymbol{n})_x, (\boldsymbol{\tau}\boldsymbol{n})_y, \boldsymbol{u}\cdot(\boldsymbol{\tau}\boldsymbol{n}) + k\nabla T\cdot\boldsymbol{n}]^T.$$

3 Meshes, discretization, limiter, and boundary conditions

CGNS triangles and quadrilaterals are imported as a cell-centred unstructured mesh. With each face normal pointing out of an owned cell, assembly adds $[(\mathbf{F}^i - \mathbf{F}^v)\cdot\boldsymbol{n}]|S|$ to that cell residual and its negative to the other adjacent owned row. Ghost states are exchanged before reconstruction, so this sign convention is retained across partition cuts. Least-squares primitive gradients provide piecewise-linear reconstruction and a Barth–Jespersen limiter bounds inviscid reconstructed values by local neighbours. A pressure-ratio sensor locally reverts the *inviscid face reconstruction* to first order across shocks. The current implementation stores viscous velocity/temperature gradients separately from the optional reconstruction-continuation scale, but the pressure-ratio sensor also limits those stored gradients in strong-compression cells. This coupling is retained in the submitted results and is stated as a limitation rather than claimed to be an independently second-order viscous shock treatment. Density/temperature floors and rejected-update recovery preserve physical primitive states. Rusanov/local-Lax–Friedrichs supplies the inviscid numerical flux. Farfield states use a characteristic trace, slip walls enforce zero normal velocity, and no-slip adiabatic walls impose $u_w = v_w = 0$ and $\partial T/\partial n = 0$; the wall-gradient construction enforces the zero velocity trace on non-orthogonal cells.

4 Implicit and transient integration

Steady cases use local pseudo-time continuation with the controls recorded in the submitted commands and metadata. The frozen residual is corrected by a rank-local LU–SGS/block-Jacobi Rusanov preconditioner; safeguarded matrix-free corrections are accepted only when a fully reassembled nonlinear residual decreases. Each reported outer residual and force is globally reduced. The Re 200 calculation uses BDF1 startup followed by BDF2,

$$A(3U^{n+1} - 4U^n + U^{n-1})/(2\Delta t) + R(U^{n+1}) = 0.$$

Both BDF history states remain frozen while the inner physical-time residual is solved and advance only after the step is accepted. The generated Re 200 statistics use the actual sampled force history, not pseudo-time iterations.

5 MPI decomposition, output, and reproducibility

METIS partitions the face-adjacent cell graph. Each rank owns cells plus a one-ring ghost layer and exchanges states and gradients only with neighbours using nonblocking point-to-point MPI. Full mesh and conservative-state replication are not used during iterations. Rank-local VTU pieces, a PVTU index, and a complete multi-piece VTU file preserve the final distributed field. A fresh checkout can reproduce the build and a representative run with

```
cmake -S solver -B solver/build
cmake --build solver/build -j
mpirun -np 8 solver/build/cfd_solver solve \
  --case cfd_solver_agentic_benchmark/inputs/cases/naca0012_m080_inviscid.json \
  --output solver/results/naca0012_m080_inviscid --report-level full
python3 solver/tools/generate_report.py \
  --report solver/report <submitted-output-directories>
cd solver/report && pdflatex report.tex && pdflatex report.tex
```

The exact command and output path for every submitted run are recorded in `run_manifest.csv`; field and CSV provenance is recorded in `figure_manifest.csv`.

6 Results, validation, and limitations

6.1 Run status and residual evidence

Table 1 is generated directly from `run_manifest.csv`; residual orders are recomputed from the first and final global L_2 rows, not copied from status text. For each steady case, generation also rejects a history below the residual target in the immutable input named by the recorded command.

Ref.	Run	Case	ranks	steps	time	residual orders	wall s	status
R1	naca0012_m015_inviscid_np8	naca0012_m015_inviscid	8	2495	172.41	4.574	7321.4	converged
R2	naca0012_m080_inviscid_np8	naca0012_m080_inviscid	8	1164	411.11	4.001	1278.2	converged
R3	naca0012_m200_inviscid_np8	naca0012_m200_inviscid	8	704	155.72	3.337	3603.9	converged
R4	naca0012_m015_laminar_re5000_np8	naca0012_m015_laminar_re5000	8	3966	291.39	4.06	12031	converged
R5	naca0012_m080_laminar_re5000_np8	naca0012_m080_laminar_re5000	8	645	36.583	5.511	5125.7	converged
R6	naca0012_m200_laminar_re5000_np8	naca0012_m200_laminar_re5000	8	787	143.46	3.642	801.41	converged
R7	cylinder_m010_laminar_re20_np8	cylinder_m010_laminar_re20	8	2621	306.51	5.842	6835.4	converged
R8	cylinder_m010_laminar_re200_np16	cylinder_m010_laminar_re200	16	30000	300	3.979	19322	statistically_periodic
R9	naca0012_m015_inviscid_np2	naca0012_m015_inviscid	2	2741	181.89	4.485	7850.8	converged
R10	cylinder_m010_laminar_re20_np2	cylinder_m010_laminar_re20	2	2675	311.94	5.824	6992.1	converged

Table 1: Run status and global residual reduction from submitted histories.

6.2 Input-case, mesh, and boundary provenance

For every output, the generator parses `run_status.command` with Python `shlex`, resolves its unique `--case` and `--output` arguments, and rejects a command whose output path or JSON case identifier does not match the submitted directory. The mesh basename is also checked against submitted metadata. Thus the tags below are evidence from the immutable command-selected input, rather than post-processed metadata. The exact JSON path remains in each manifest command; R-codes map to Table 1.

Ref.	mesh	physics	cells	faces	boundary-tag mapping
R1	NACA0012_H2.cgns	inviscid, M=0.15	20816	36498	bc-2 → farfield; bc-4 → slip wall
R2	NACA0012_H2.cgns	inviscid, M=0.8	20816	36498	bc-2 → farfield; bc-4 → slip wall
R3	NACA0012_H2.cgns	inviscid, M=2	20816	36498	bc-2 → farfield; bc-4 → slip wall
R4	NACA0012_H2.cgns	laminar, M=0.15, Re=5000	20816	36498	bc-2 → farfield; bc-4 → no-slip adiabatic wall
R5	NACA0012_H2.cgns	laminar, M=0.8, Re=5000	20816	36498	bc-2 → farfield; bc-4 → no-slip adiabatic wall
R6	NACA0012_H2.cgns	laminar, M=2, Re=5000	20816	36498	bc-2 → farfield; bc-4 → no-slip adiabatic wall
R7	CylinderB1.cgns	laminar, M=0.1, Re=20	10185	20180	WALL → no-slip adiabatic wall; FAR → farfield
R8	CylinderB1.cgns	laminar, M=0.1, Re=200	10185	20180	WALL → no-slip adiabatic wall; FAR → farfield
R9	NACA0012_H2.cgns	inviscid, M=0.15	20816	36498	bc-2 → farfield; bc-4 → slip wall
R10	CylinderB1.cgns	laminar, M=0.1, Re=20	10185	20180	WALL → no-slip adiabatic wall; FAR → farfield

Table 2: Case, mesh, and boundary-tag evidence from the exact JSON named by each submitted command.

6.3 Recorded numerical controls

Control cells are deliberately source-marked: [M] is a submitted top-level metadata field, [A] an applied-control field, [R] a serialized submitted `run_control`, and [I] the immutable input JSON named by `--case`. The [I] fallback is used only when a legacy output omitted that control; no value is inferred from a plot, defaulted silently, or written back into metadata. [–] means that neither source specifies the field.

Ref.	sources	integrator	type	horizon
R1	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=20000 [I]
R2	M+A+R	steady local pseudo-time block LU–SGS [M]	steady [R]	max steps=30000 [R]
R3	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=40000 [I]
R4	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=40000 [I]
R5	M+A+R	steady local pseudo-time block LU–SGS [M]	steady [R]	max steps=40000 [R]
R6	M+A+R	steady local pseudo-time block LU–SGS [M]	steady [R]	max steps=50000 [R]
R7	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=30000 [I]
R8	M+I	BDF2 or trapezoidal [M]	transient [I]	dt=0.01 [I]; tf=300 [I]
R9	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=20000 [I]
R10	M+I	steady pseudo-time block-Jacobi [M]	steady [I]	max steps=30000 [I]

Table 3: Integration and horizon controls with their serialized provenance.

Ref.	CFL initial	CFL max	outer target	inner target	Rusanov scale
R1	1 [I]	100 [I]	4 [I]	0.01 [M]	not specified [–]
R2	1 [R]	100 [R]	4 [R]	0.01 [M]	1 [A]
R3	0.2 [I]	50 [I]	3 [I]	0.01 [M]	not specified [–]
R4	1 [I]	100 [I]	4 [I]	0.01 [M]	not specified [–]
R5	0.5 [R]	80 [R]	4 [R]	0.01 [M]	1 [A]
R6	0.2 [R]	50 [R]	3 [R]	0.01 [M]	1 [A]
R7	1 [I]	100 [I]	5 [I]	0.01 [M]	not specified [–]
R8	1 [I]	1 [I]	not specified [–]	0.001 [M]	1 [I]
R9	1 [I]	100 [I]	4 [I]	0.01 [M]	not specified [–]
R10	1 [I]	100 [I]	5 [I]	0.01 [M]	not specified [–]

Table 4: CFL, residual-target, and dissipation controls with per-cell provenance markers.

6.4 Forces and convergence qualification

The final-state force values below are read from the final row of each submitted `forces.csv`; pressure and viscous contributions remain separate. A steady row is admitted only after the residual history reaches the target specified in its command-selected input JSON.

Ref.	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	qualification
R1	0.0021014	-0.00016132	3.296e-05	0.0021014	0	steady target met
R2	0.008572	0.00091907	8.594e-05	0.008572	0	steady target met
R3	0.092006	0.00053819	0.0003313	0.092006	0	steady target met
R4	0.075121	-0.00045793	-0.000327	0.022457	0.052664	steady target met
R5	0.078843	-0.0034961	-0.0009781	0.044125	0.034718	steady target met
R6	0.044548	6.1989e-05	-1.64e-05	0.018123	0.026425	steady target met
R7	2.0372	-0.0003392	-1.038e-05	1.2387	0.79853	steady target met
R8	1.1524	-0.087456	-0.002553	0.88761	0.26475	statistically periodic
R9	0.001209	8.3203e-05	4.77e-05	0.001209	0	steady target met
R10	2.037	-0.00032495	-9.729e-06	1.2386	0.79842	steady target met

Table 5: Final force coefficients and convergence qualification; R-codes map to Table 1.

6.5 Case-by-case interpretation

Each paragraph below refers to the corresponding submitted histories and final field. These plots support qualitative flow interpretation, not an unperformed grid-convergence or experimental-validation claim.

naca0012_m015_inviscid_np8. This inviscid, M=0.15 submitted run is classified converged with 4.574 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.00210136$ and $C_L = -0.000161318$. Figure 1 and Figure 2 show the convergence/force history; Figure 3 and Figure 5 and Figure 7 show the wall-pressure and near-body/wake field evidence. The low-Mach result is reported with the same field and surface evidence, without extrapolating it to an incompressible reference solution.

naca0012_m080_inviscid_np8. This inviscid, M=0.8 submitted run is classified converged with 4.001 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.00857203$ and $C_L = 0.000919066$. Figure 8 and Figure 9 show the convergence/force history; Figure 10 and Figure 12 and Figure 14 show the wall-pressure and near-body/wake field evidence. The finite-Mach field panels make the compression and expansion structure inspectable without treating the contour itself as an external validation result.

naca0012_m200_inviscid_np8. This inviscid, M=2 submitted run is classified converged with 3.337 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.0920059$ and $C_L = 0.000538188$. Figure 15 and Figure 16 show the convergence/force history; Figure 17 and Figure 19 and Figure 21 show the wall-pressure and near-body/wake field evidence. The supplied Mach and pressure fields are the direct evidence used for the shock/expansion-dominated finite-Mach pattern; no reference-data error or mesh-refinement uncertainty is asserted.

naca0012_m015_laminar_re5000_np8. This laminar, M=0.15, Re=5000 submitted run is classified converged with 4.06 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.0751211$ and $C_L = -0.000457933$. Figure 22 and Figure 23 show the convergence/force history; Figure 24 and Figure 27 and Figure 29 show the wall-pressure and near-body/wake field evidence. The low-Mach result is reported with the same field and surface evidence, without extrapolating it to an incompressible reference solution. The reported tangential wall stress is plotted in Figure 25.

naca0012_m080_laminar_re5000_np8. This laminar, M=0.8, Re=5000 submitted run is classified converged with 5.511 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.0788425$ and $C_L = -0.00349607$. Figure 30 and Figure 31 show the convergence/force history; Figure 32 and Figure 35 and Figure 37 show the wall-pressure and near-body/wake field evidence. The finite-Mach field panels make the compression and expansion structure inspectable without treating the contour itself as an external validation result. The reported tangential wall stress is plotted in Figure 33.

naca0012_m200_laminar_re5000_np8. This laminar, M=2, Re=5000 submitted run is classified converged with 3.642 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.0445481$ and $C_L = 6.19888e{-}05$. Figure 38 and Figure 39 show the convergence/force history; Figure 40 and Figure 43 and Figure 45 show the wall-pressure and near-body/wake field evidence. The supplied Mach and pressure fields are the direct evidence used for the shock/expansion-dominated finite-Mach pattern; no reference-data error or mesh-refinement uncertainty is asserted. The reported tangential wall stress is plotted in Figure 41.

cylinder_m010_laminar_re20_np8. This laminar, M=0.1, Re=20 submitted run is classified converged with 5.842 orders of visible L_2 reduction. Its final coefficients are $C_D = 2.03721$ and $C_L = -0.000339201$. Figure 46 and Figure 47 show the convergence/force history; Figure 48 and Figure 51 and Figure 53 show the wall-pressure and near-body/wake field evidence. The reported tangential wall stress is plotted in Figure 49. Its low-Re wake assessment is based on the final submitted field and the settled force history.

cylinder_m010_laminar_re200_np16. This laminar, M=0.1, Re=200 submitted run is classified statistically_periodic with 3.979 orders of visible L_2 reduction. Its final coefficients are $C_D = 1.15235$ and $C_L = -0.0874561$. Figure 54 and Figure 55 show the convergence/force history; Figure 56 and Figure 59 and Figure 61 show the wall-pressure and near-body/wake field evidence. The reported tangential wall stress is plotted in Figure 57. The post-transient wake diagnostic is Figure 62.

naca0012_m015_inviscid_np2. This inviscid, M=0.15 submitted run is classified converged with 4.485 orders of visible L_2 reduction. Its final coefficients are $C_D = 0.00120899$ and $C_L = 8.32032e{-}05$. Figure 63 and Figure 64 show the convergence/force history; Figure 65 and Figure 67 and Figure 69 show the wall-pressure and near-body/wake field evidence. The low-Mach result is reported with the same field and surface evidence, without extrapolating it to an incompressible reference solution.

cylinder_m010_laminar_re20_np2. This laminar, M=0.1, Re=20 submitted run is classified converged with 5.824 orders of visible L_2 reduction. Its final coefficients are $C_D = 2.03699$ and $C_L = -0.000324947$. Figure 70 and Figure 71 show the convergence/force history; Figure 72 and Figure 75 and Figure 77 show the wall-pressure and near-body/wake field evidence. The reported tangential wall stress is plotted in Figure 73. Its low-Re wake assessment is based on the final submitted field and the settled force history.

6.6 Partition diagnostics and MPI comparison

The first table summarizes the actual per-rank diagnostics written by the solver. The following long table retains every owned/ghost count, boundary-face count, neighbour count, and summed send/receive cell count rather than collapsing communication evidence to an average. 'Neighbour graph edges' is half the directed neighbour count in the submitted partition diagnostics. In the detailed table, R1, R2, and so on identify the corresponding row order in Table 1.

Ref.	owned min–max	ghost min–max	balance	avg. neigh.	edges	sent	received
R1	2550–2627	47–201	1.0096	3	12	831	831
R2	2550–2627	47–201	1.0096	3	12	831	831
R3	2550–2627	47–201	1.0096	3	12	831	831
R4	2550–2627	47–201	1.0096	3	12	831	831
R5	2550–2627	47–201	1.0096	3	12	831	831
R6	2550–2627	47–201	1.0096	3	12	831	831
R7	1262–1281	99–122	1.0062	4.25	17	915	915
R8	625–651	68–111	1.0227	5	40	1383	1383
R9	10405–10411	119–120	1.0003	1	1	239	239
R10	5090–5095	100–104	1.0005	1	1	204	204

Table 6: METIS partition balance, ghost-layer, neighbour, and communication totals; R-codes map to Table 1.

Table 7: Per-rank partition diagnostics as emitted by the submitted solver.

run ref.	rank	owned	ghost	boundary faces	neighbours	send cells	receive cells
R1	0	2550	88	70	2	88	88
R1	1	2627	91	63	3	91	91
R1	2	2623	108	56	3	109	108
R1	3	2610	92	64	3	91	92
R1	4	2557	47	80	1	47	47
R1	5	2613	201	20	7	201	201
R1	6	2612	96	60	2	96	96
R1	7	2624	108	71	3	108	108
R2	0	2550	88	70	2	88	88
R2	1	2627	91	63	3	91	91
R2	2	2623	108	56	3	109	108
R2	3	2610	92	64	3	91	92
R2	4	2557	47	80	1	47	47
R2	5	2613	201	20	7	201	201
R2	6	2612	96	60	2	96	96
R2	7	2624	108	71	3	108	108
R3	0	2550	88	70	2	88	88
R3	1	2627	91	63	3	91	91
R3	2	2623	108	56	3	109	108
R3	3	2610	92	64	3	91	92
R3	4	2557	47	80	1	47	47
R3	5	2613	201	20	7	201	201
R3	6	2612	96	60	2	96	96
R3	7	2624	108	71	3	108	108
R4	0	2550	88	70	2	88	88
R4	1	2627	91	63	3	91	91
R4	2	2623	108	56	3	109	108
R4	3	2610	92	64	3	91	92
R4	4	2557	47	80	1	47	47
R4	5	2613	201	20	7	201	201
R4	6	2612	96	60	2	96	96
R4	7	2624	108	71	3	108	108
R5	0	2550	88	70	2	88	88
R5	1	2627	91	63	3	91	91
R5	2	2623	108	56	3	109	108
R5	3	2610	92	64	3	91	92
R5	4	2557	47	80	1	47	47

run ref.	rank	owned	ghost	boundary faces	neighbours	send cells	receive cells
R5	5	2613	201	20	7	201	201
R5	6	2612	96	60	2	96	96
R5	7	2624	108	71	3	108	108
R6	0	2550	88	70	2	88	88
R6	1	2627	91	63	3	91	91
R6	2	2623	108	56	3	109	108
R6	3	2610	92	64	3	91	92
R6	4	2557	47	80	1	47	47
R6	5	2613	201	20	7	201	201
R6	6	2612	96	60	2	96	96
R6	7	2624	108	71	3	108	108
R7	0	1281	120	23	5	120	120
R7	1	1262	104	25	3	103	104
R7	2	1275	113	23	4	112	113
R7	3	1273	118	29	3	118	118
R7	4	1269	99	0	3	98	99
R7	5	1276	118	20	6	120	118
R7	6	1273	121	0	4	122	121
R7	7	1276	122	0	6	122	122
R8	0	635	87	0	4	88	87
R8	1	646	75	0	5	76	75
R8	2	634	111	0	6	106	111
R8	3	635	81	20	6	82	81
R8	4	637	84	0	4	84	84
R8	5	638	91	0	6	93	91
R8	6	636	110	0	6	111	110
R8	7	634	84	0	5	84	84
R8	8	632	89	0	5	87	89
R8	9	636	75	28	4	77	75
R8	10	639	68	29	4	70	68
R8	11	625	90	0	6	89	90
R8	12	651	82	0	7	81	82
R8	13	634	72	26	3	72	72
R8	14	637	97	17	4	97	97
R8	15	636	87	0	5	86	87
R9	0	10405	120	253	1	119	120
R9	1	10411	119	231	1	120	119
R10	0	5095	100	100	1	104	100
R10	1	5090	104	20	1	100	104

For naca0012_m015_inviscid, the submitted np=2 and np=8 runs give final $C_D = 0.00120899$ and 0.00210136, a 73.81% difference relative to the lower-rank value; their residual reductions are 4.48 and 4.57, and wall times 7850.83 s and 7321.43 s. This is an actual parallel consistency record, not a strong-scaling claim; material differences are reported without rounding them away. For cylinder_m010_laminar_re20, the submitted np=2 and np=8 runs give final $C_D = 2.03699$ and 2.03721, a 0.01081% difference relative to the lower-rank value; their residual reductions are 5.82 and 5.84, and wall times 6992.14 s and 6835.38 s. This is an actual parallel consistency record, not a strong-scaling claim; material differences are reported without rounding them away.

6.7 Cylinder Re 200 vortex-street analysis

Over the latter half of the sampled physical-time history ($t \in [150.01, 300]$), the mean drag is 1.15175, mean lift is -0.00449455, lift RMS is 0.236634, and lift half-range amplitude is 0.336351. Linear interpolation of 24 upward lift zero crossings gives dominant frequency 0.16333 and Strouhal number 0.16333, using the supplied unit reference length and velocity. Figure 62 is the submitted-field velocity-magnitude wake view; Figures 55 and 61 provide the force history and pressure wake context.

6.8 Figure guide and sanity checks

Every figure below is generated from the source named in its caption and recorded one-to-one in `figure_manifest.csv`. Mach and pressure contours use the actual unstructured field cells; the body/wake panels provide the prescribed near-body views. The machine-readable checks are in `sanity_checks.json`.

- naca0012_m015_inviscid: pass.
- naca0012_m080_inviscid: pass.
- naca0012_m200_inviscid: pass.
- naca0012_m015_laminar_re5000: pass.
- naca0012_m080_laminar_re5000: pass.
- naca0012_m200_laminar_re5000: pass.
- cylinder_m010_laminar_re20: pass.
- cylinder_m010_laminar_re200: pass.
- naca0012_m015_inviscid: pass.
- cylinder_m010_laminar_re20: pass.

6.9 Figures

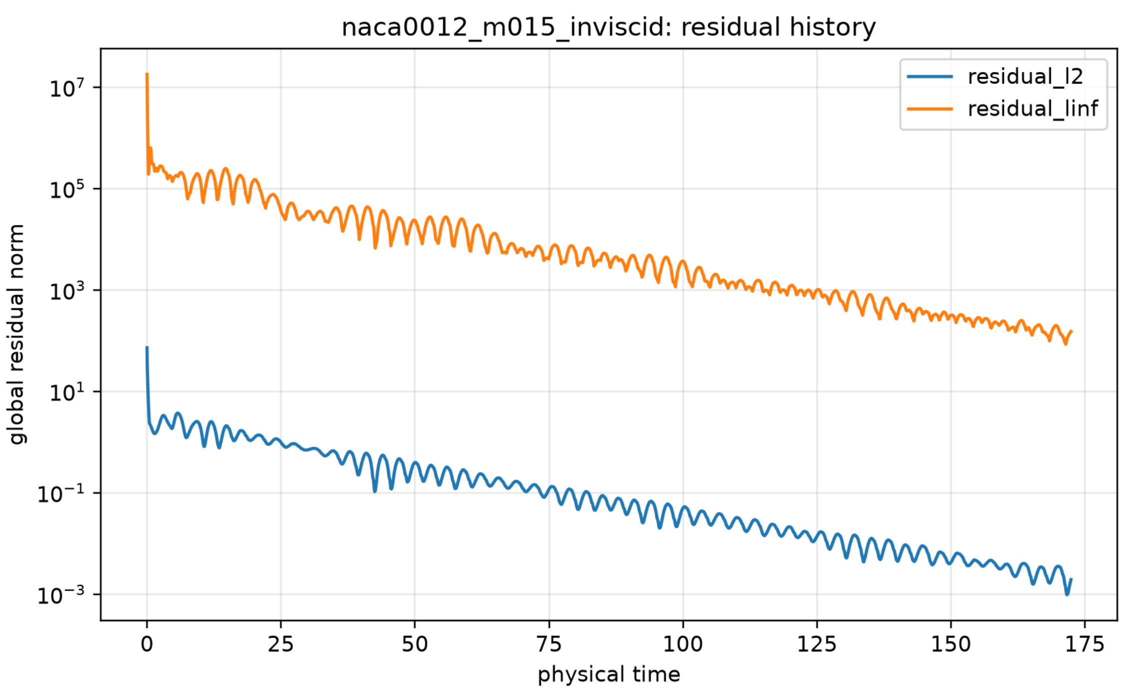


Figure 1: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

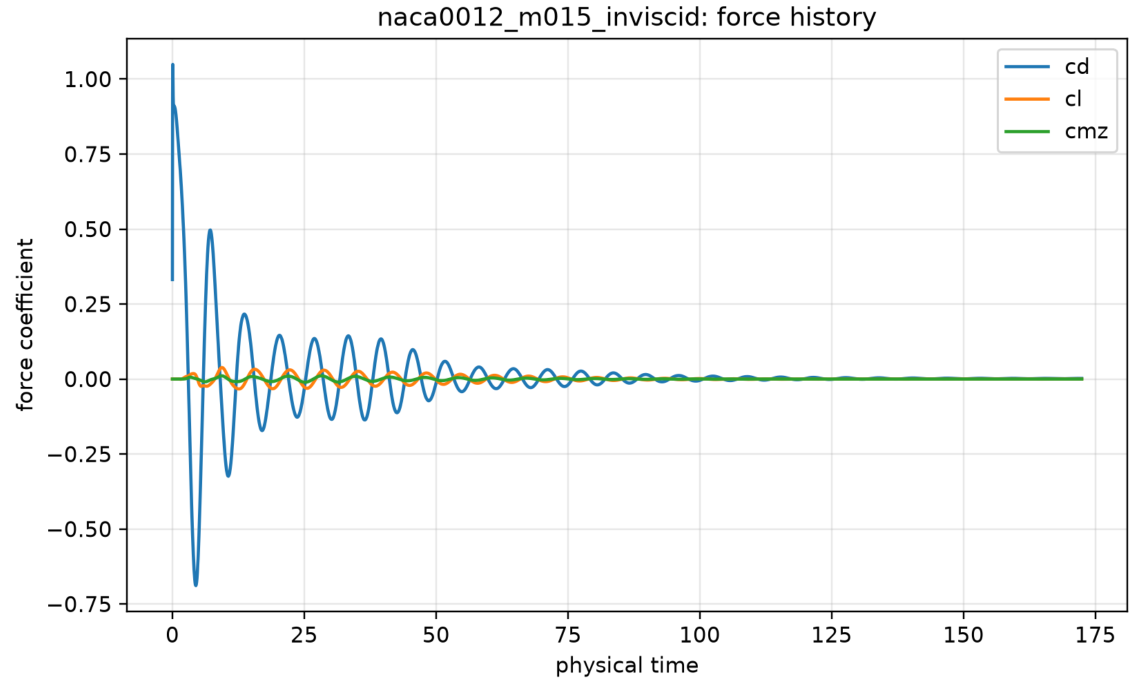


Figure 2: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

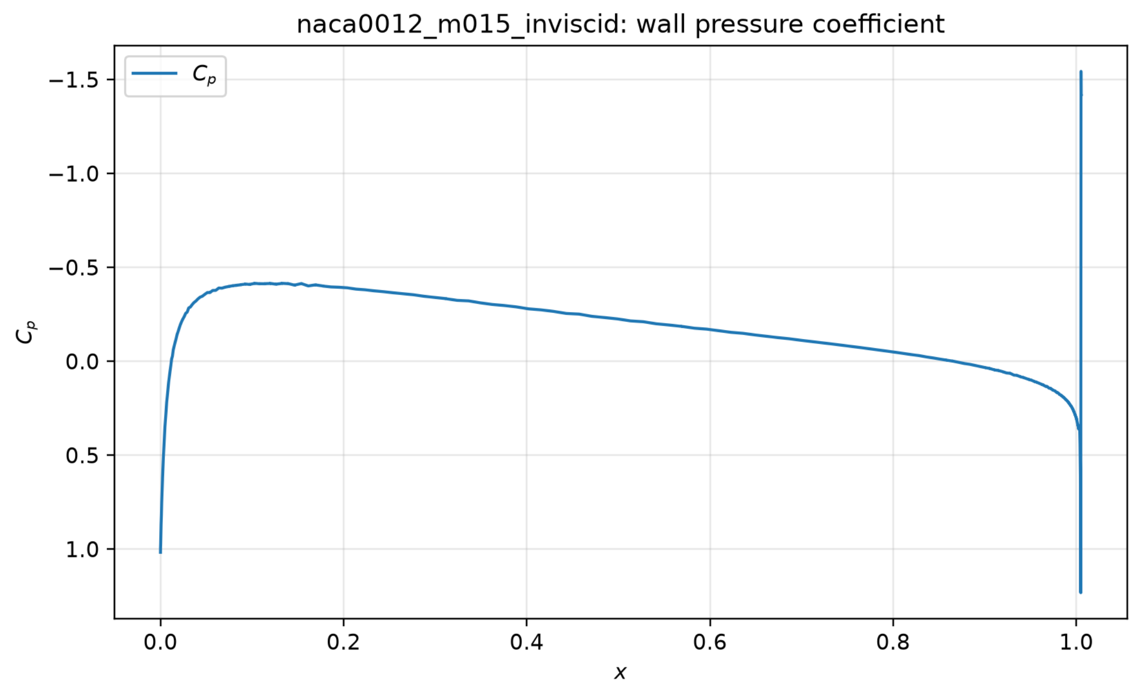


Figure 3: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

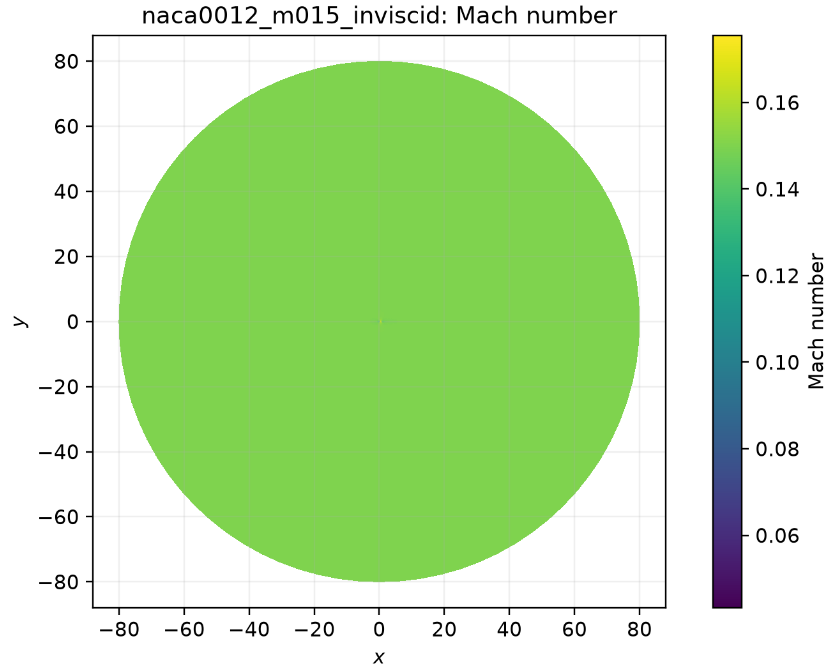


Figure 4: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

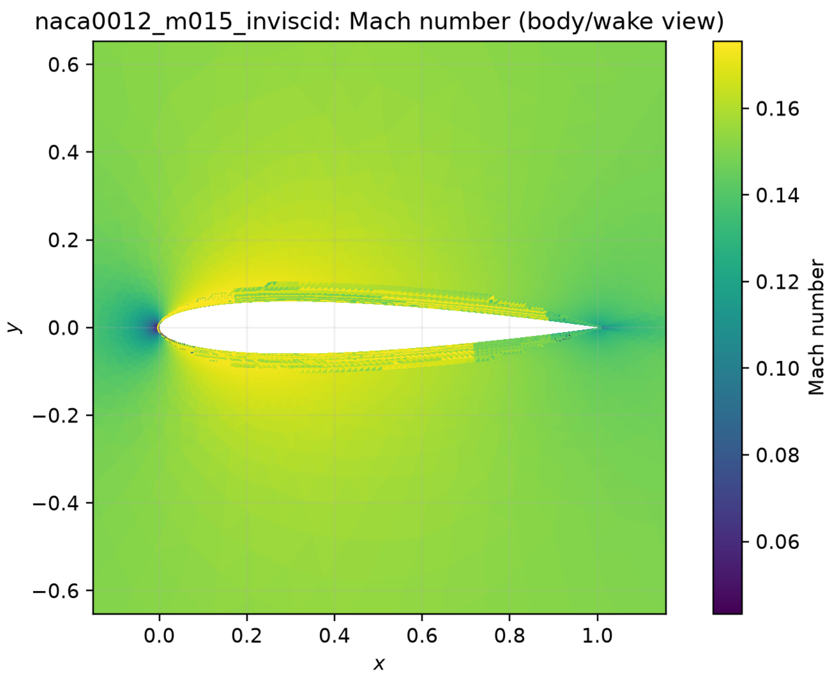


Figure 5: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

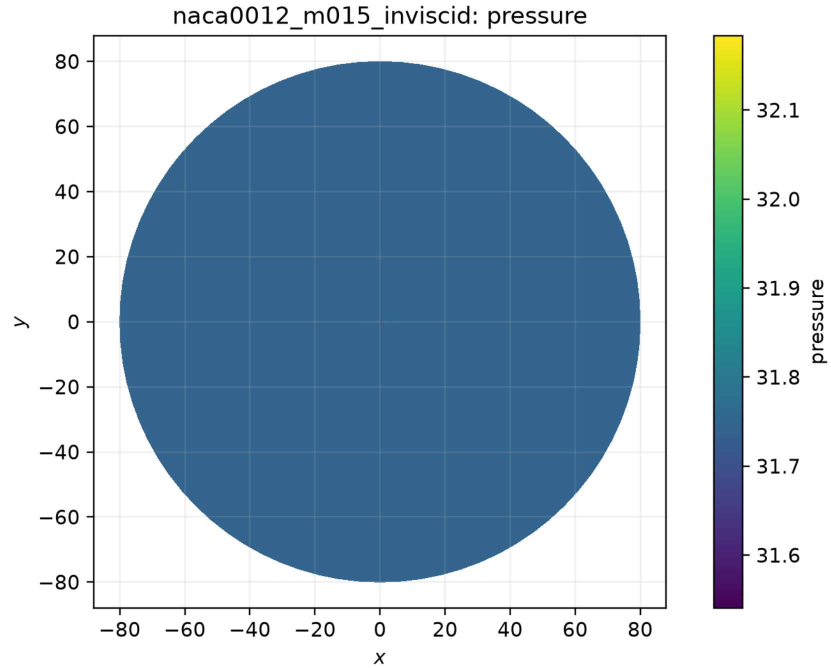


Figure 6: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

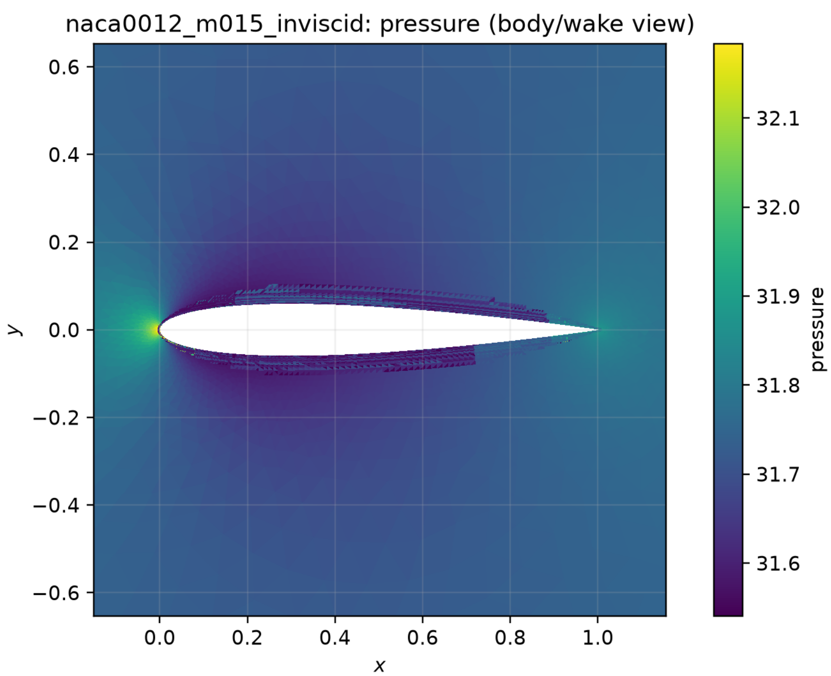


Figure 7: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

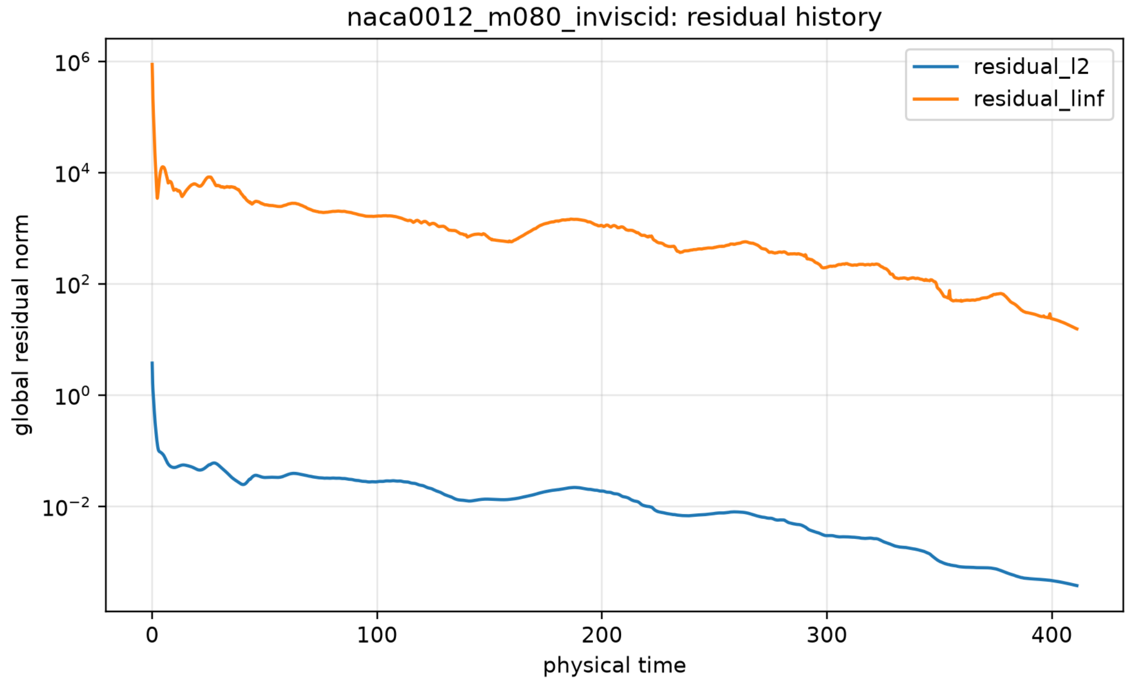


Figure 8: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

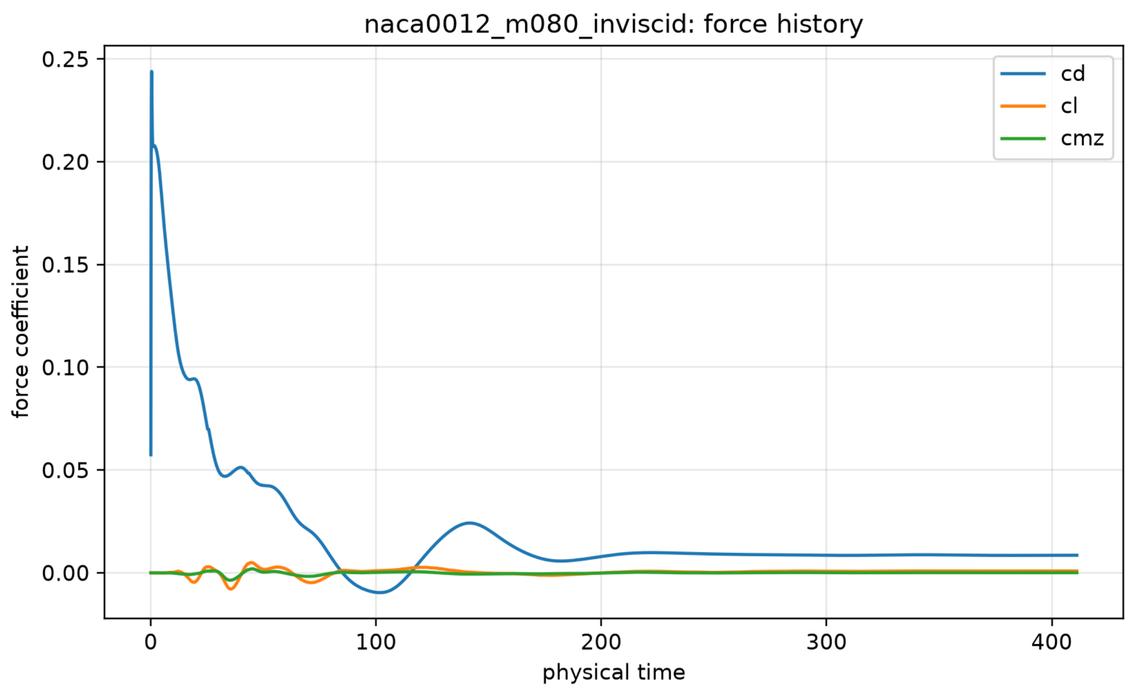


Figure 9: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

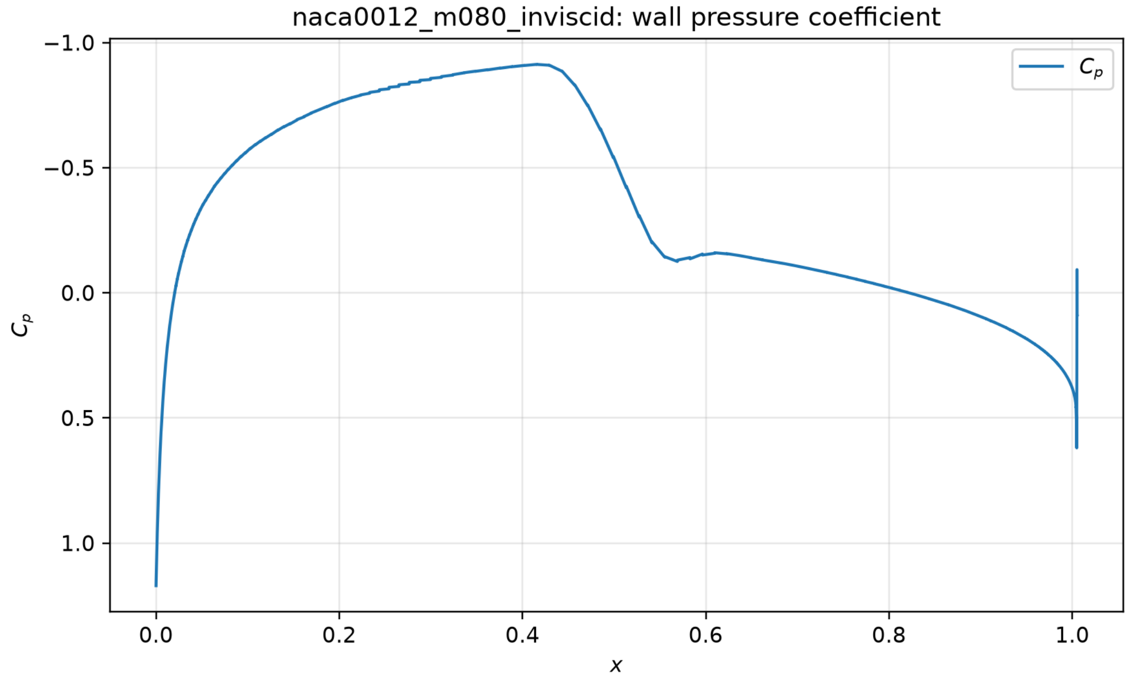


Figure 10: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

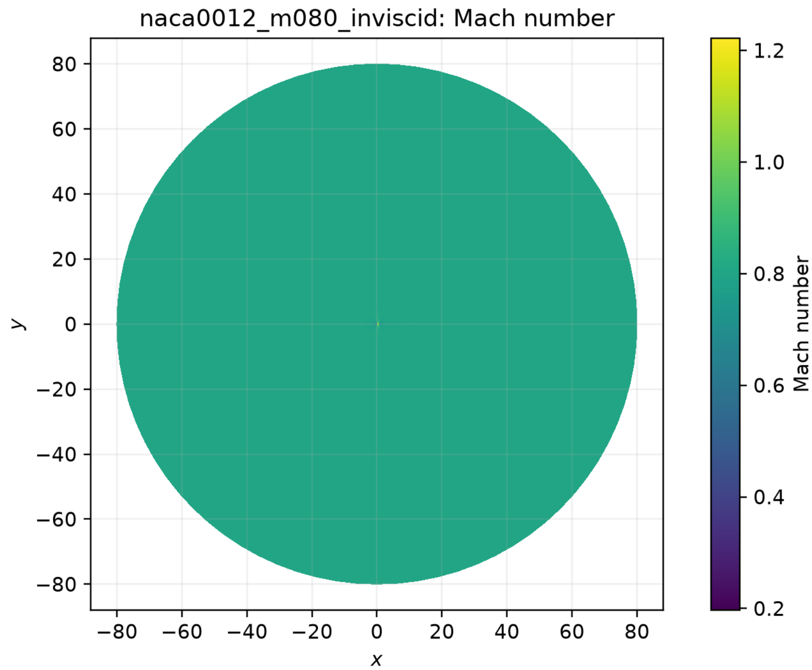


Figure 11: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

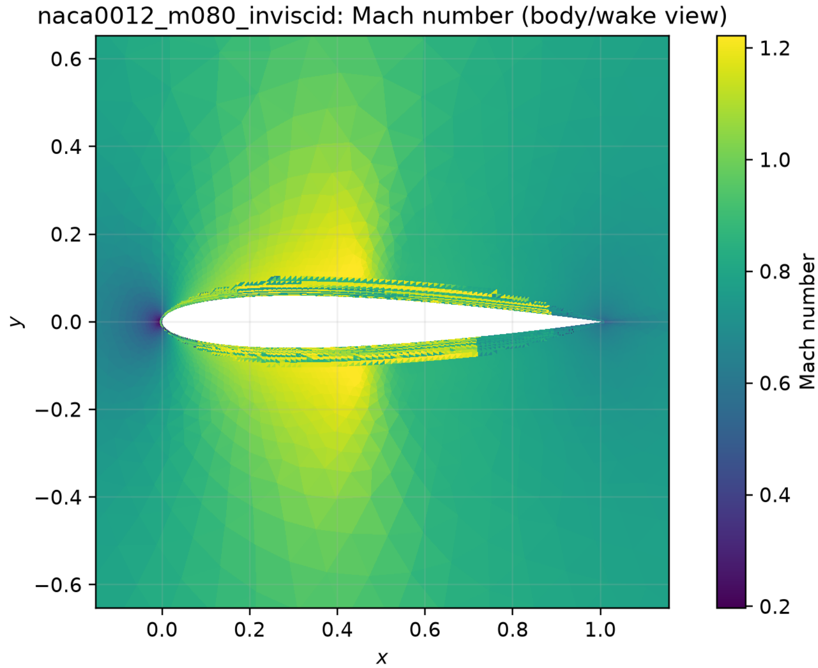


Figure 12: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

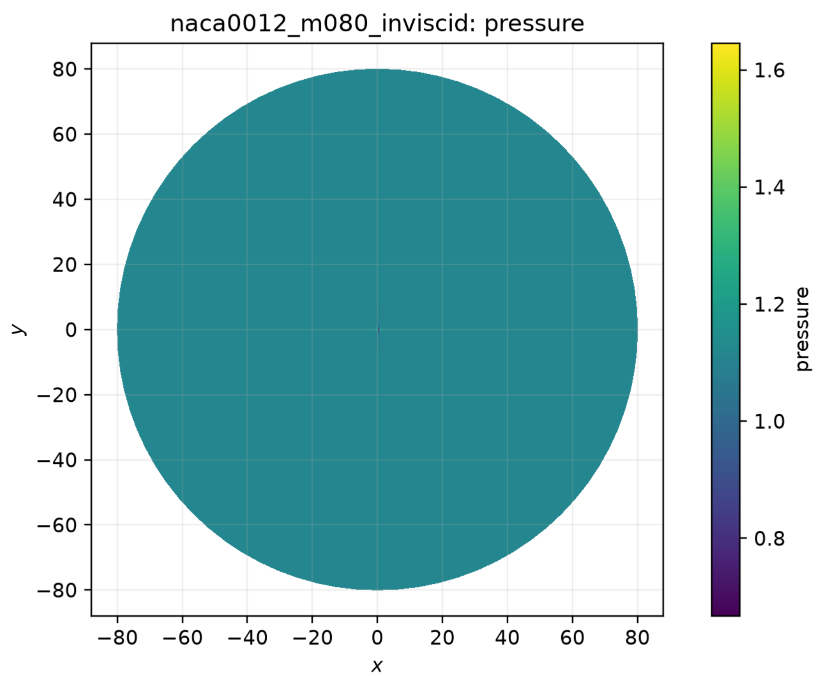


Figure 13: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

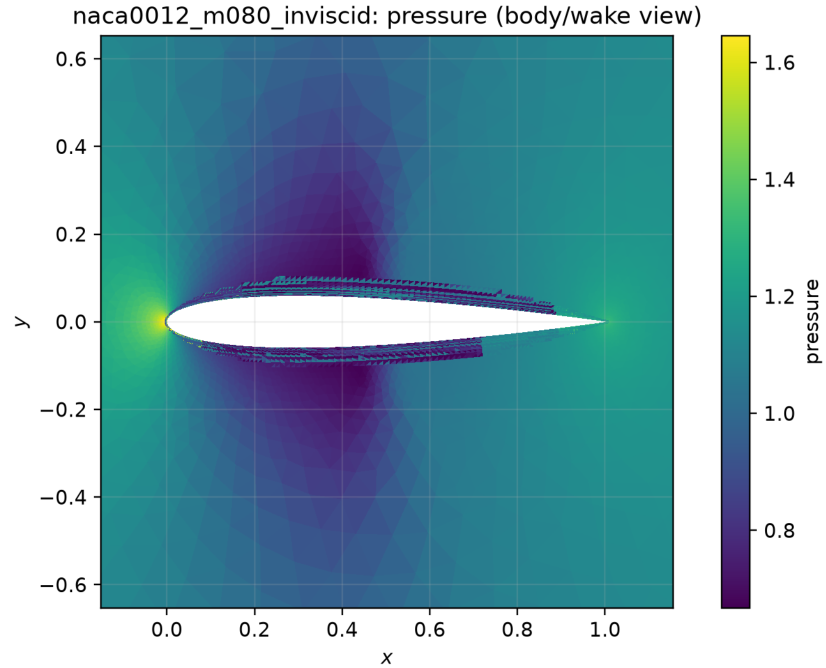


Figure 14: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

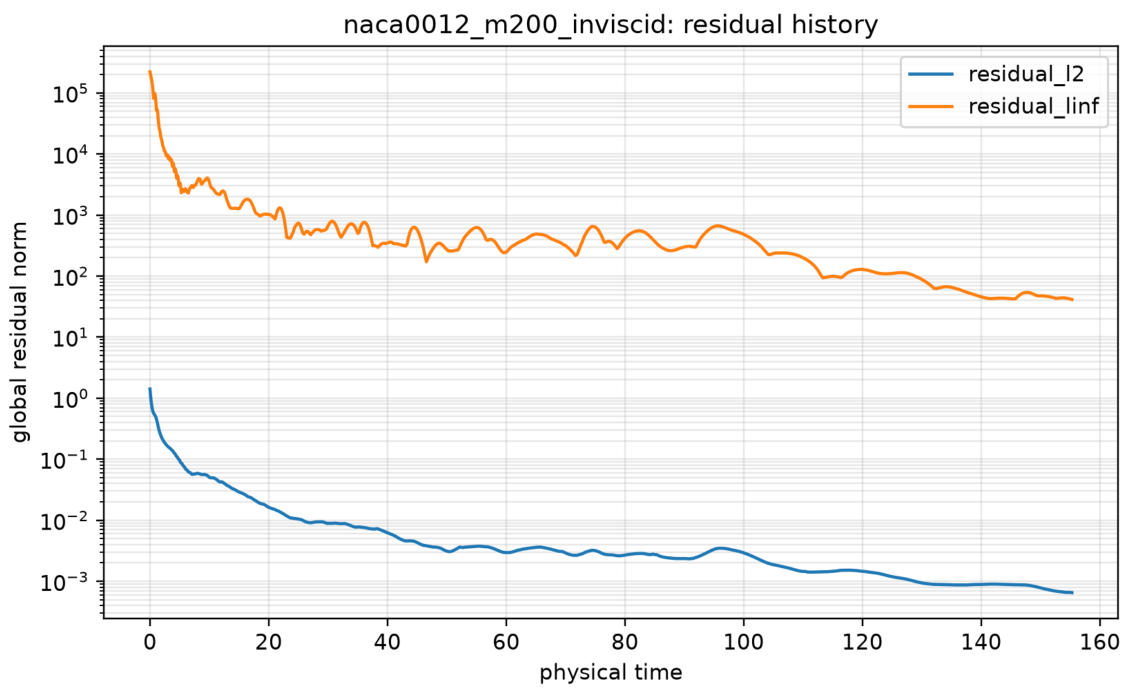


Figure 15: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

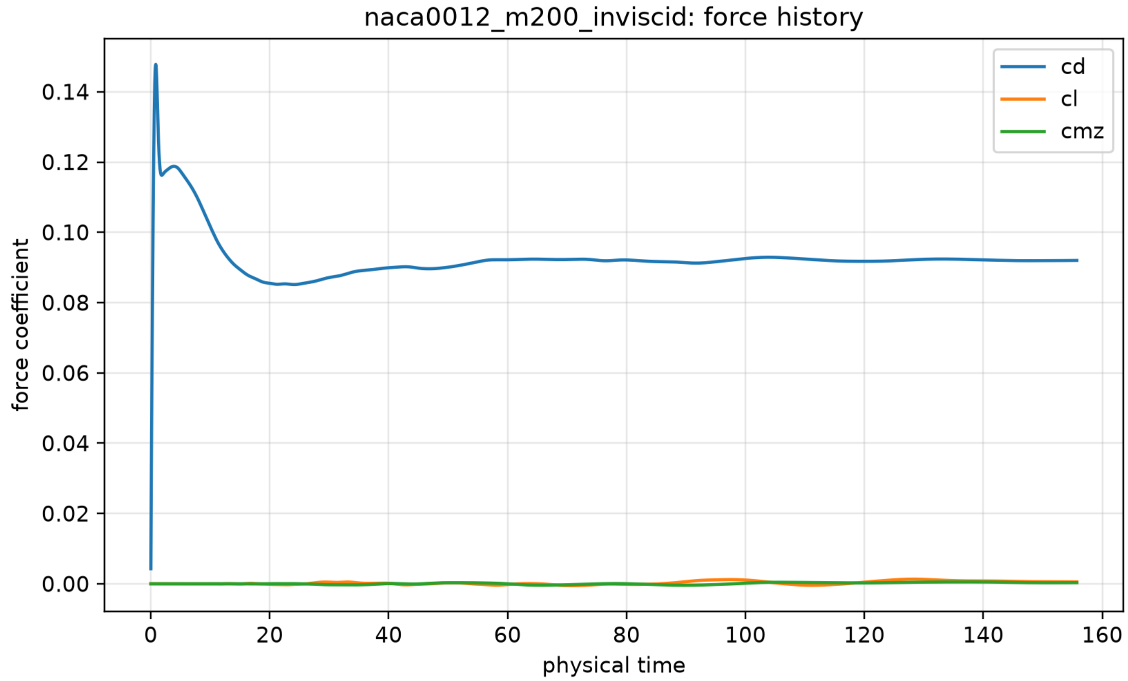


Figure 16: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

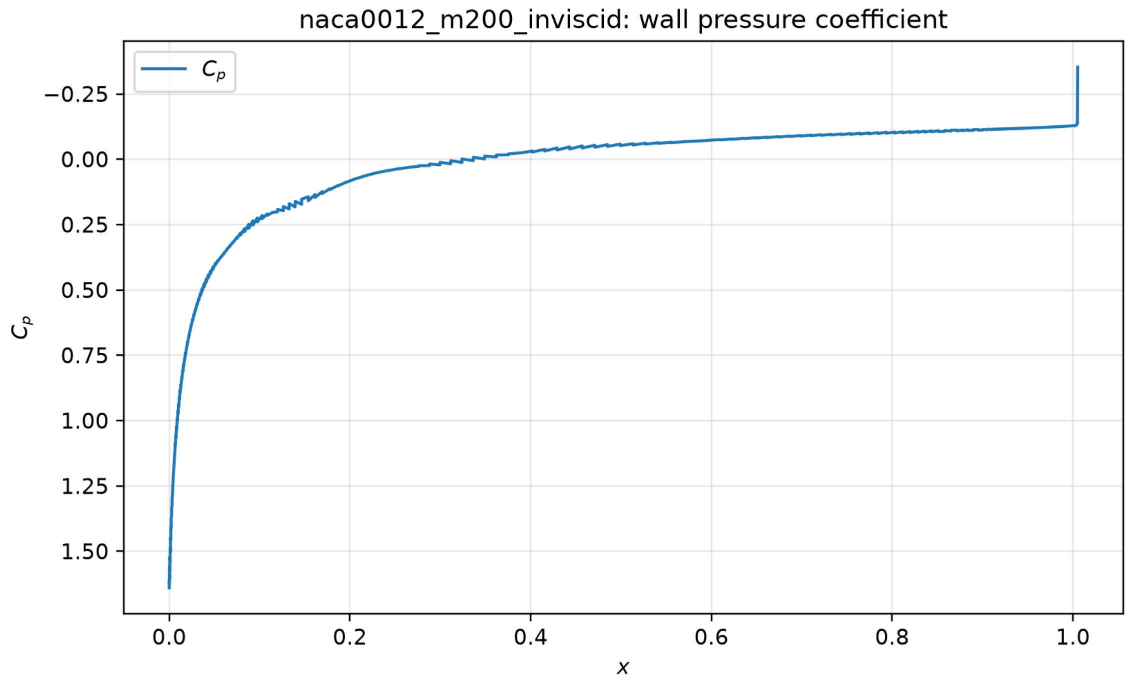


Figure 17: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

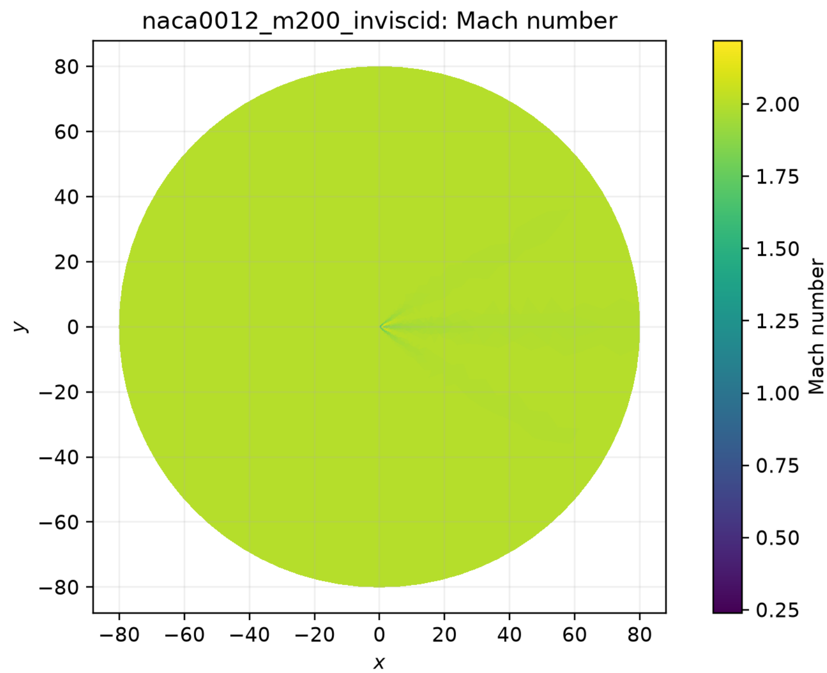


Figure 18: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

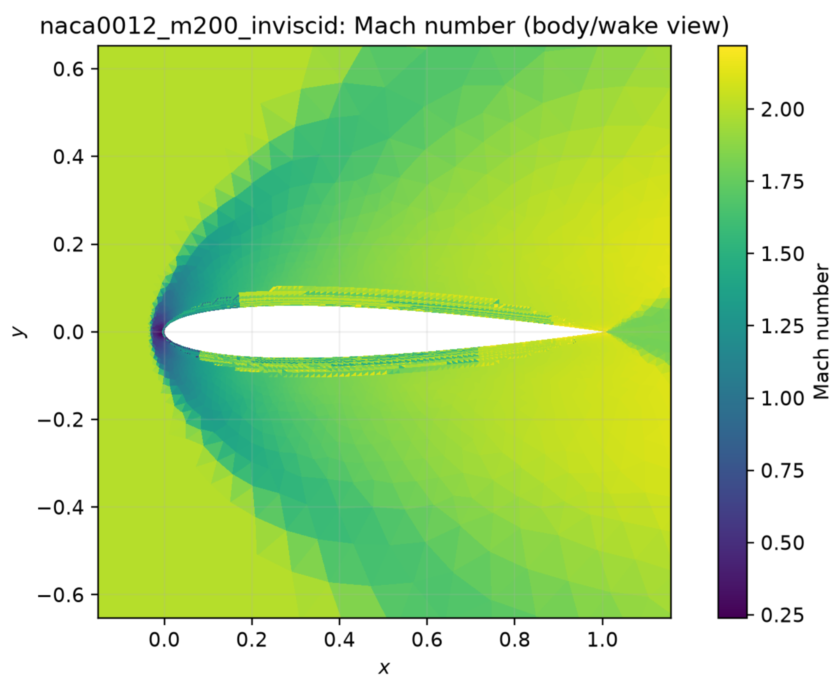
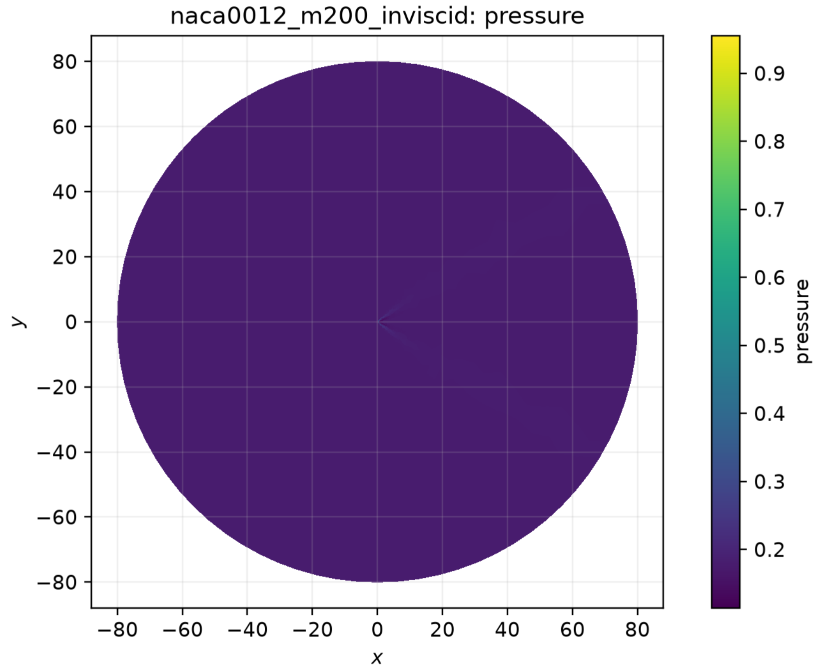


Figure 19: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.



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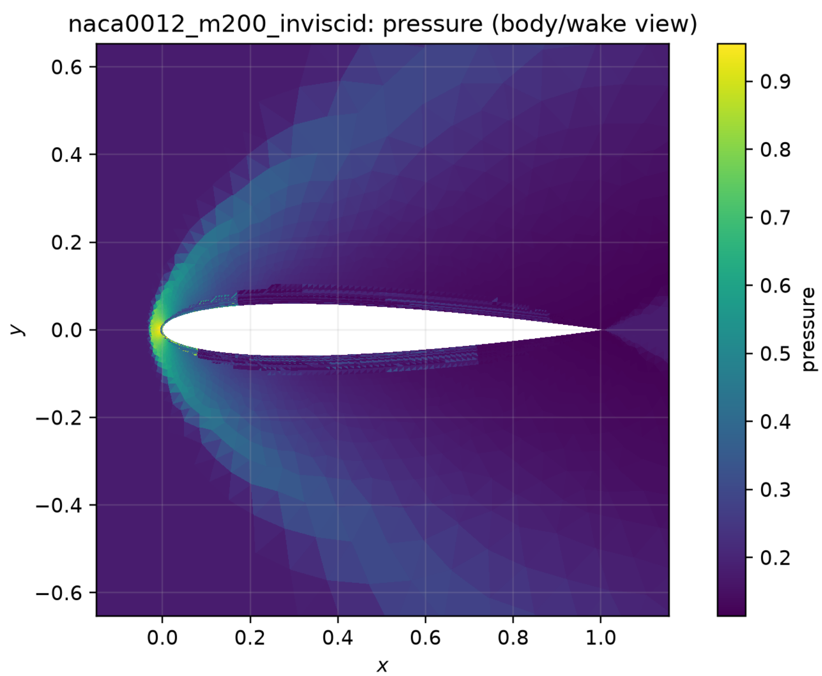

Figure 21: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

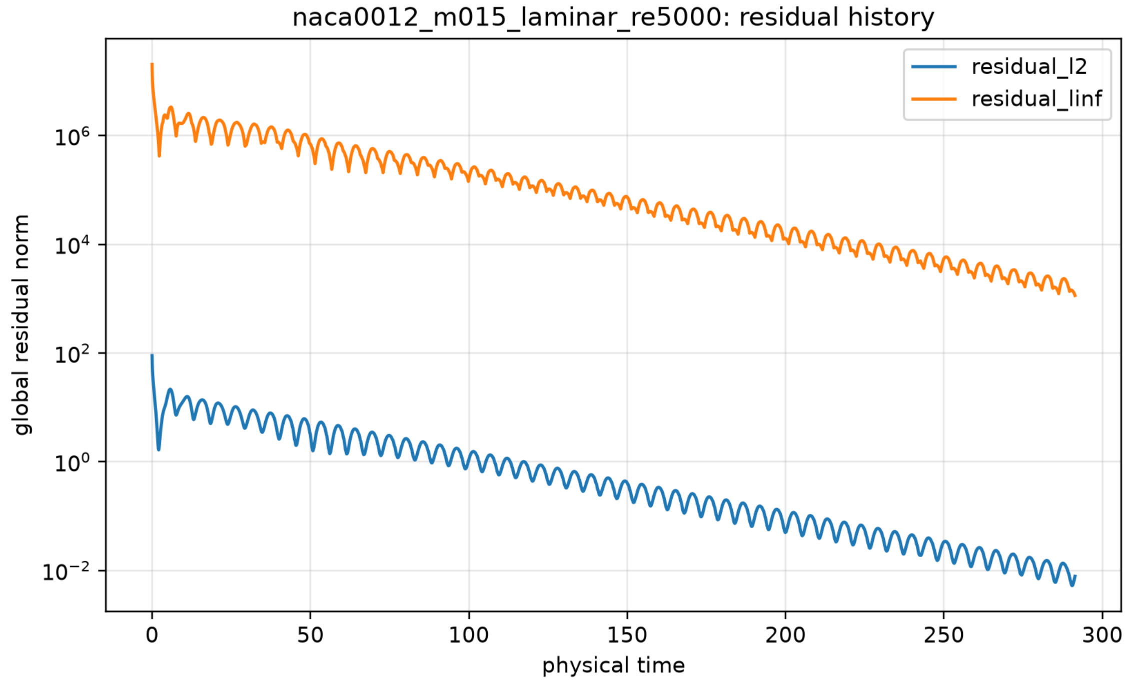

Figure 22: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

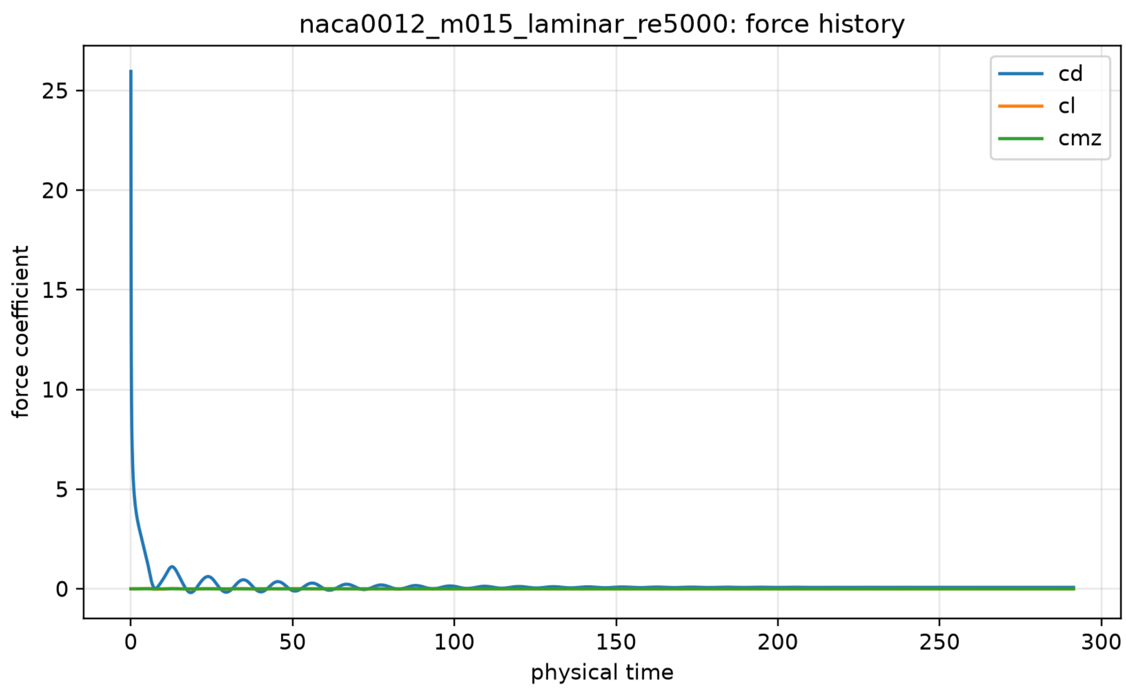

Figure 23: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

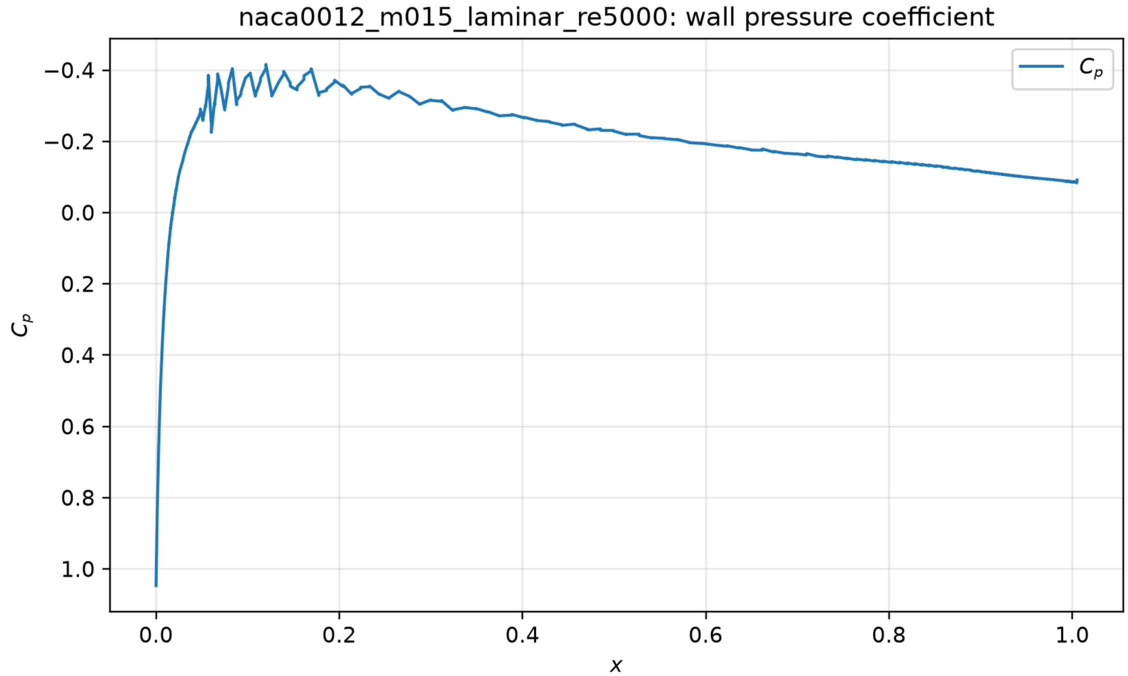

Figure 24: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

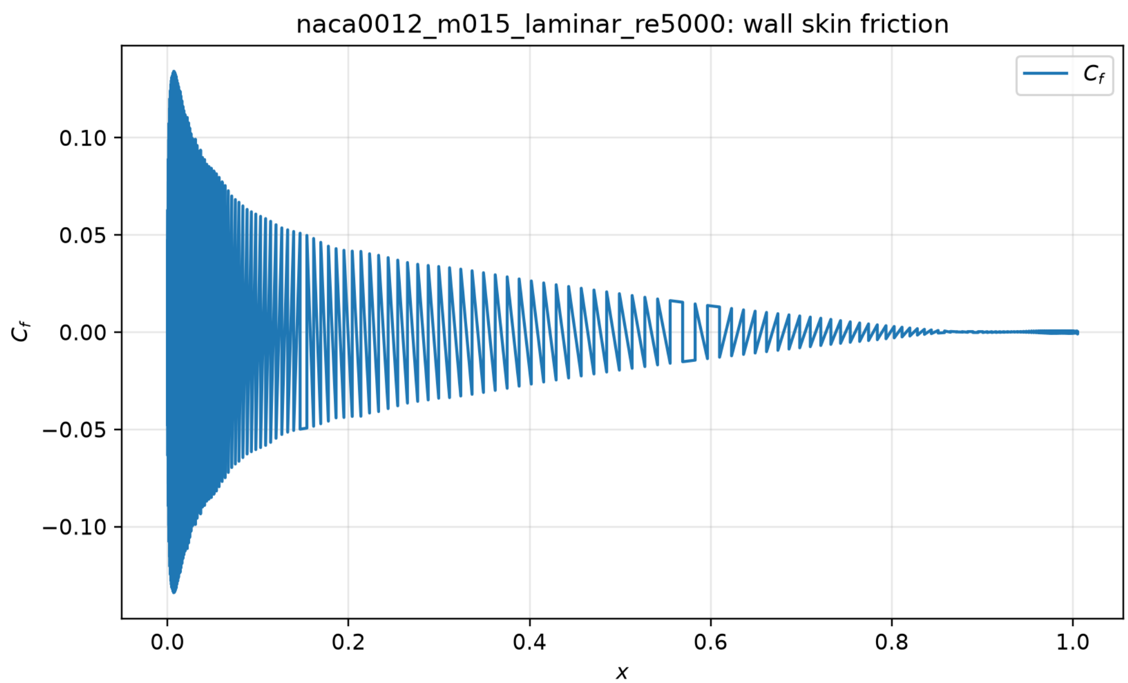

Figure 25: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

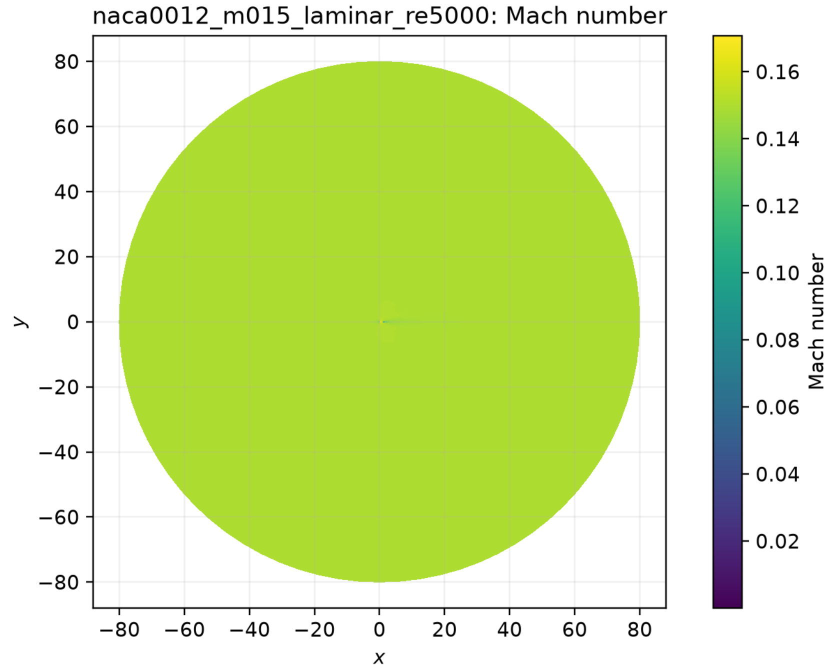

Figure 26: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

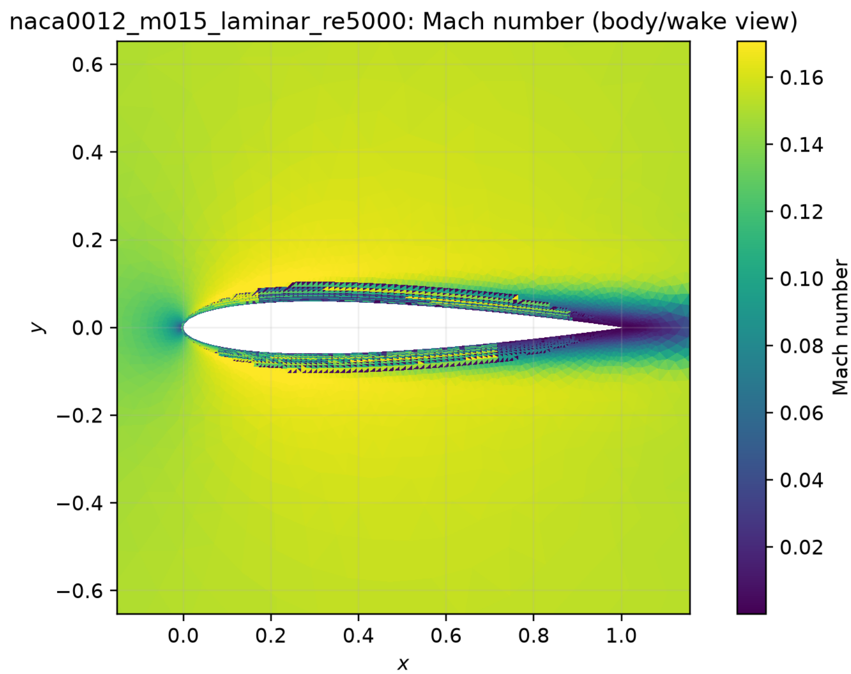

Figure 27: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

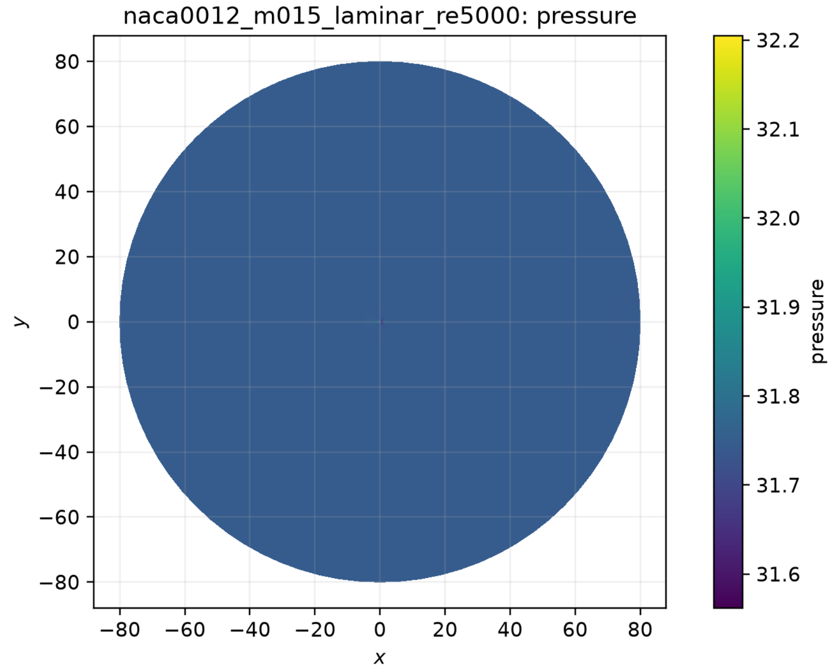

Figure 28: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

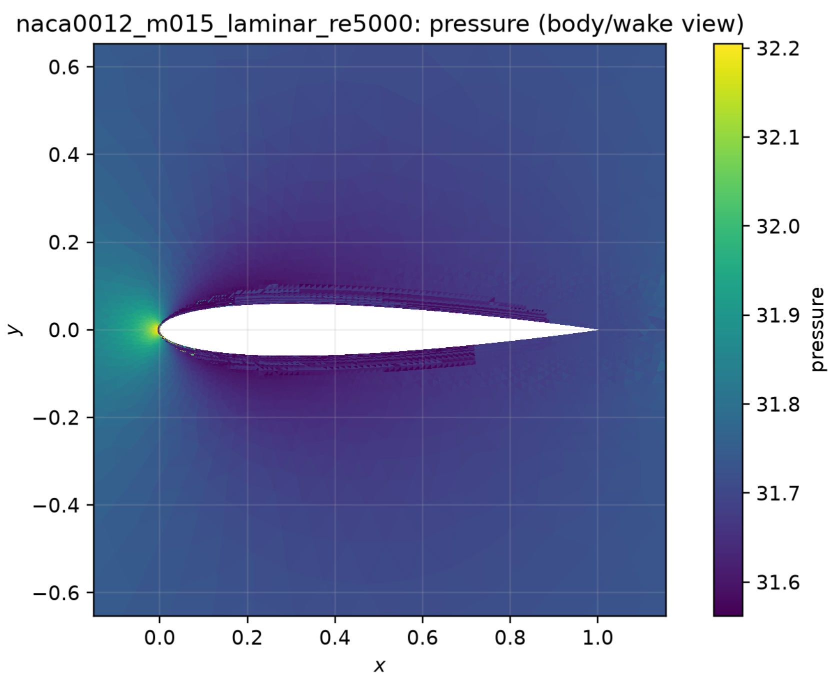

Figure 29: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

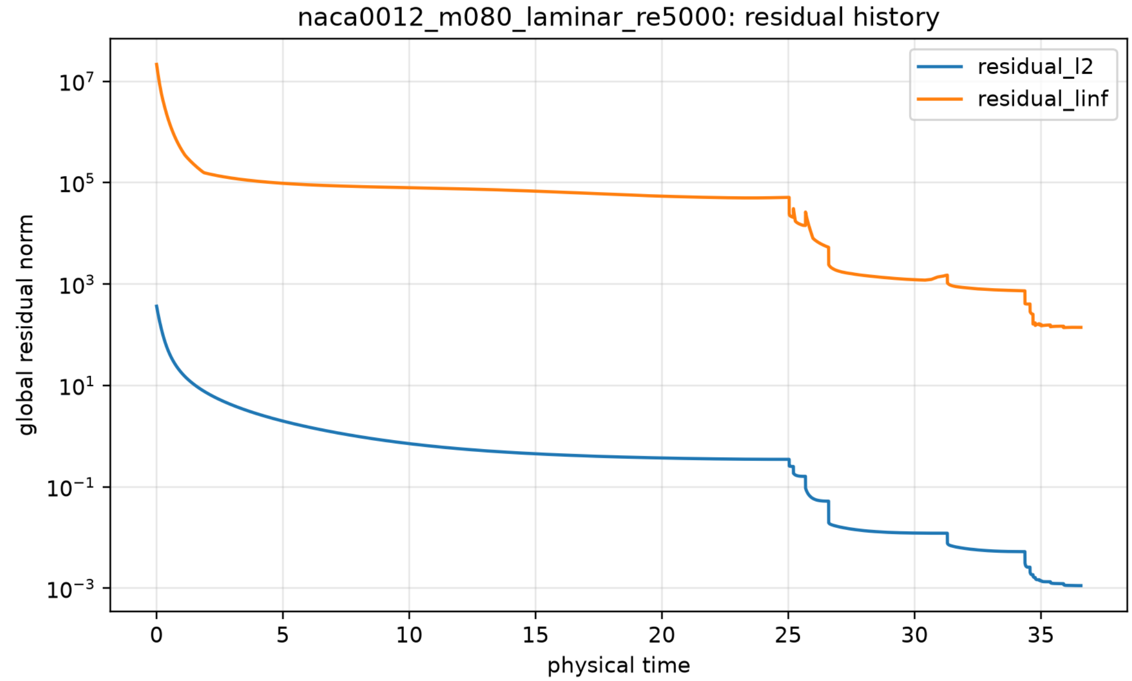

Figure 30: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

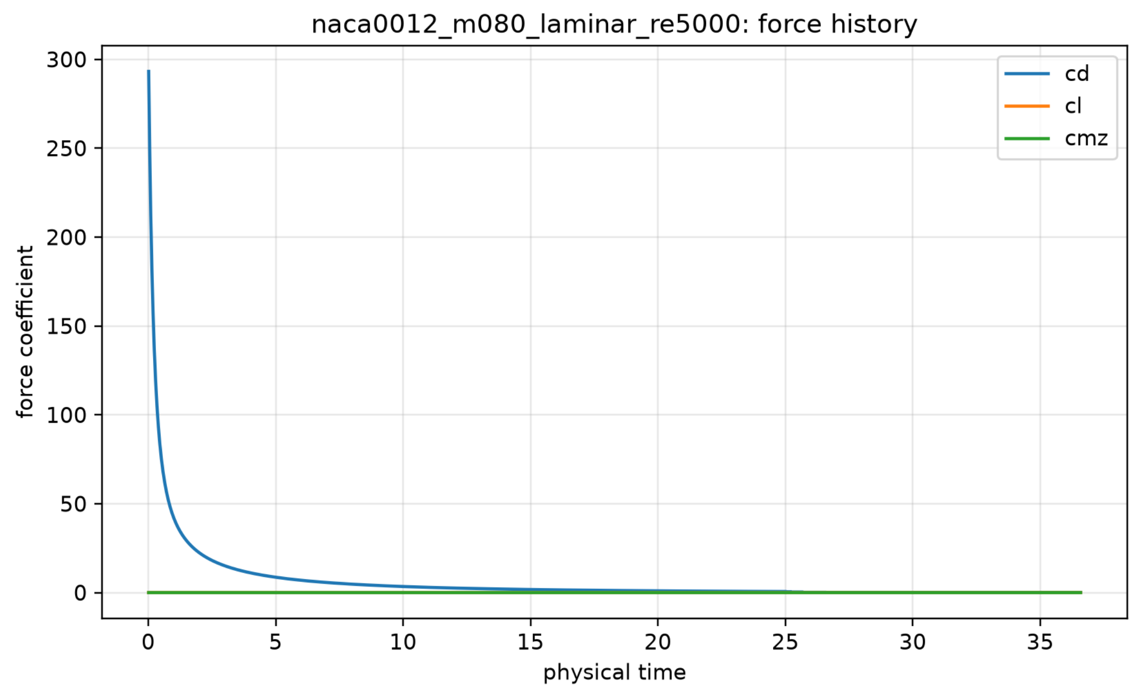

Figure 31: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

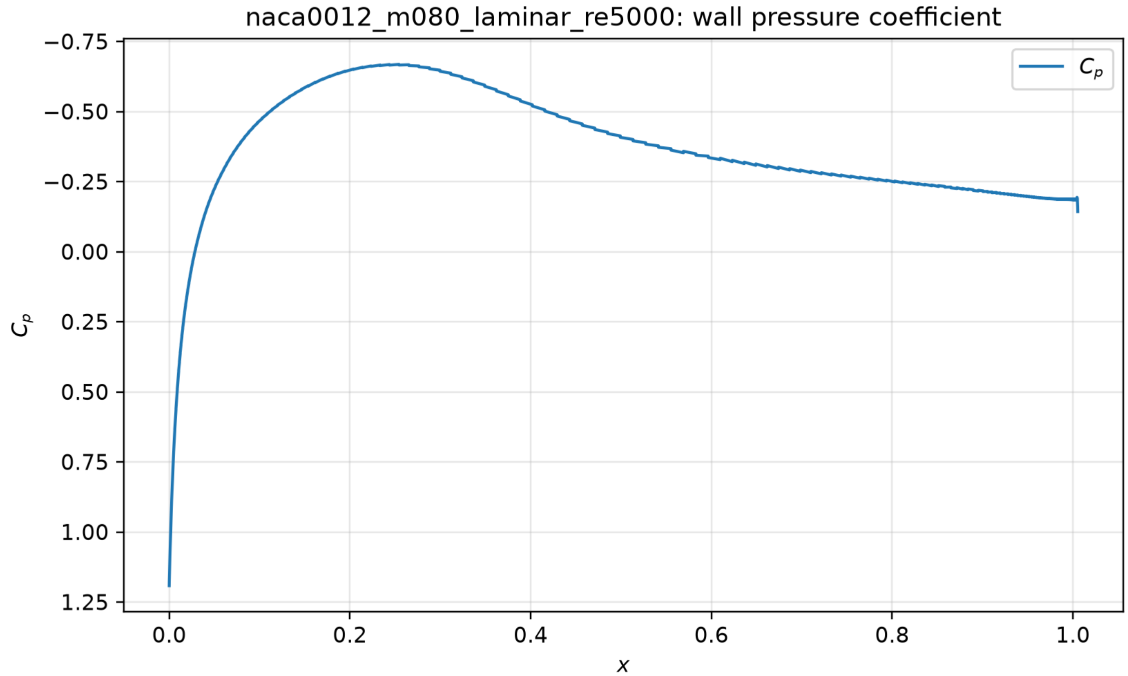

Figure 32: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

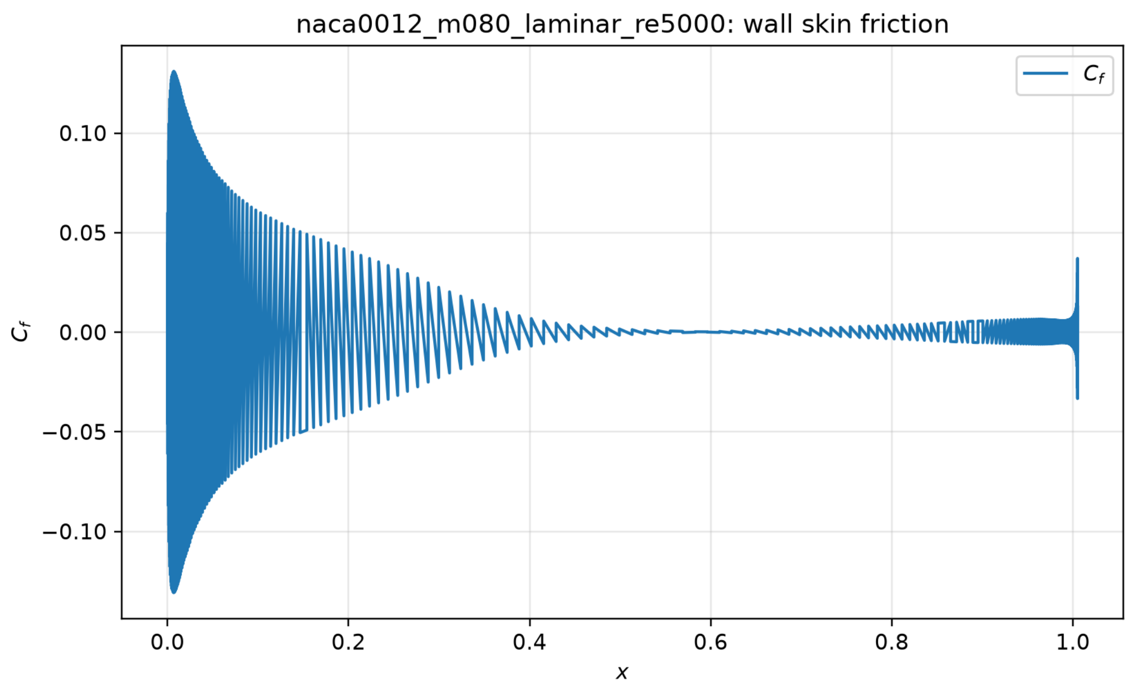

Figure 33: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

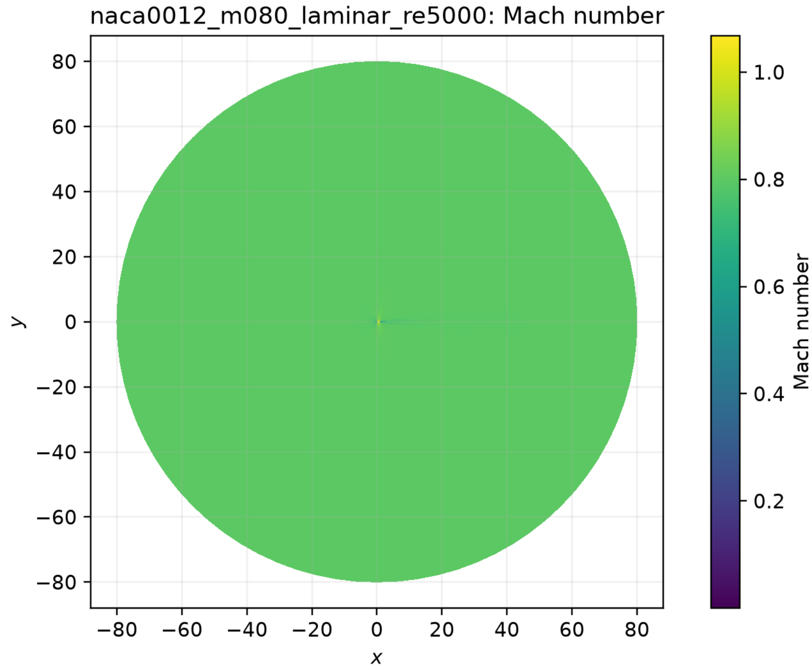

Figure 34: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

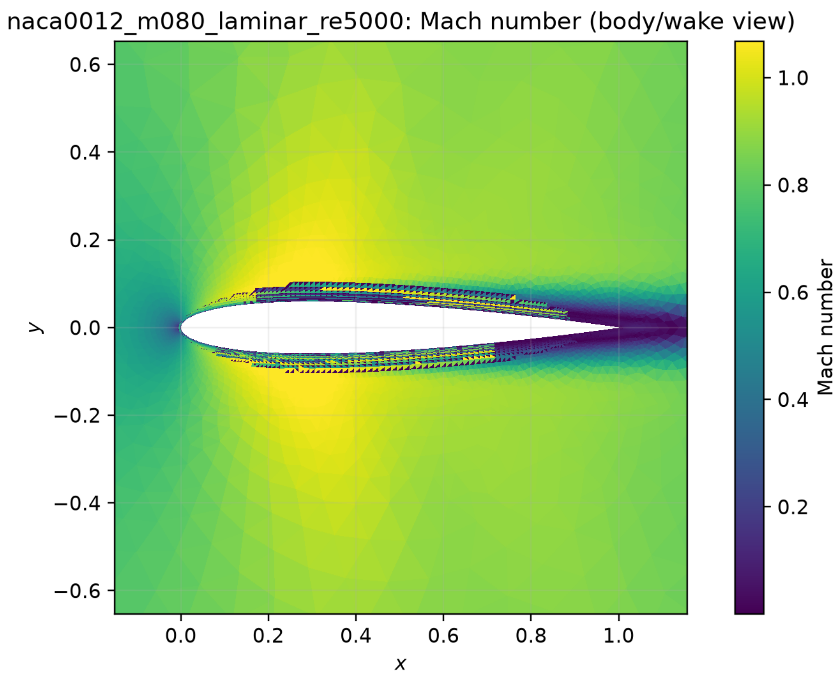

Figure 35: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

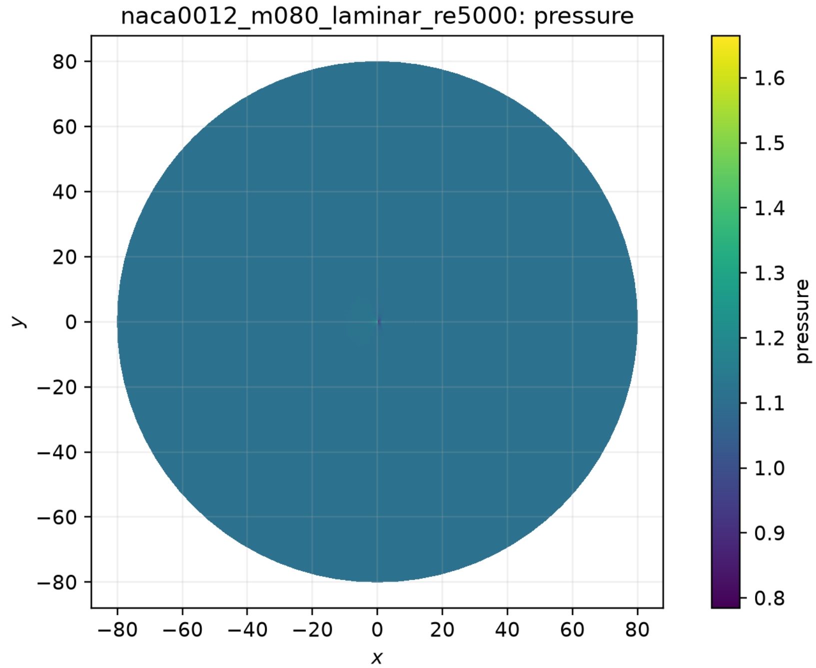

Figure 36: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

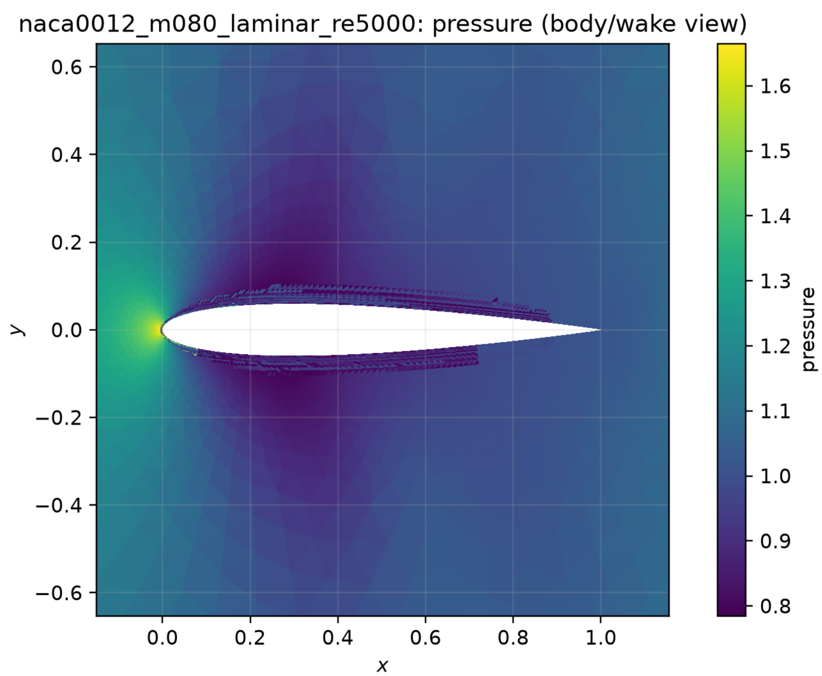

Figure 37: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

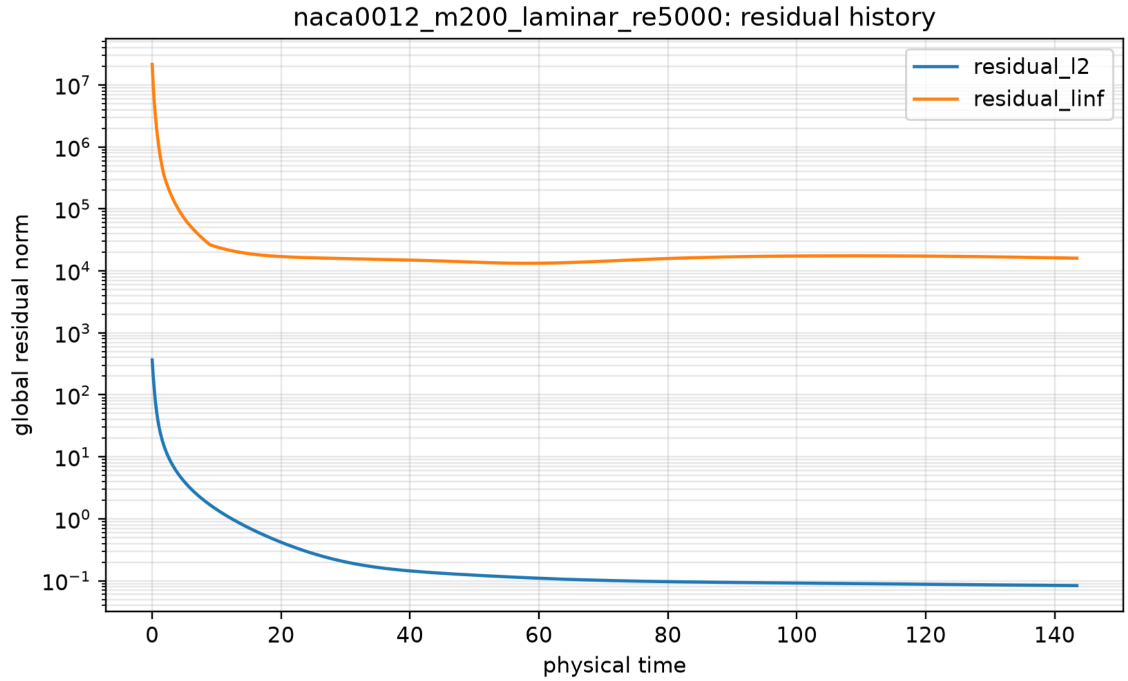

Figure 38: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

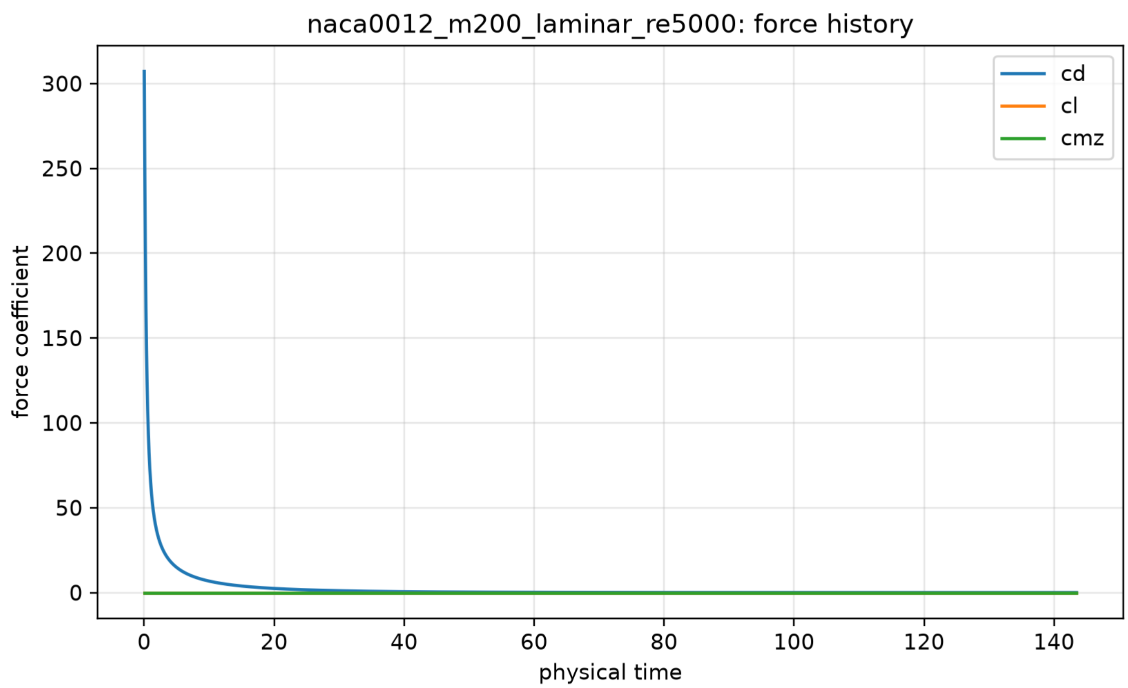

Figure 39: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

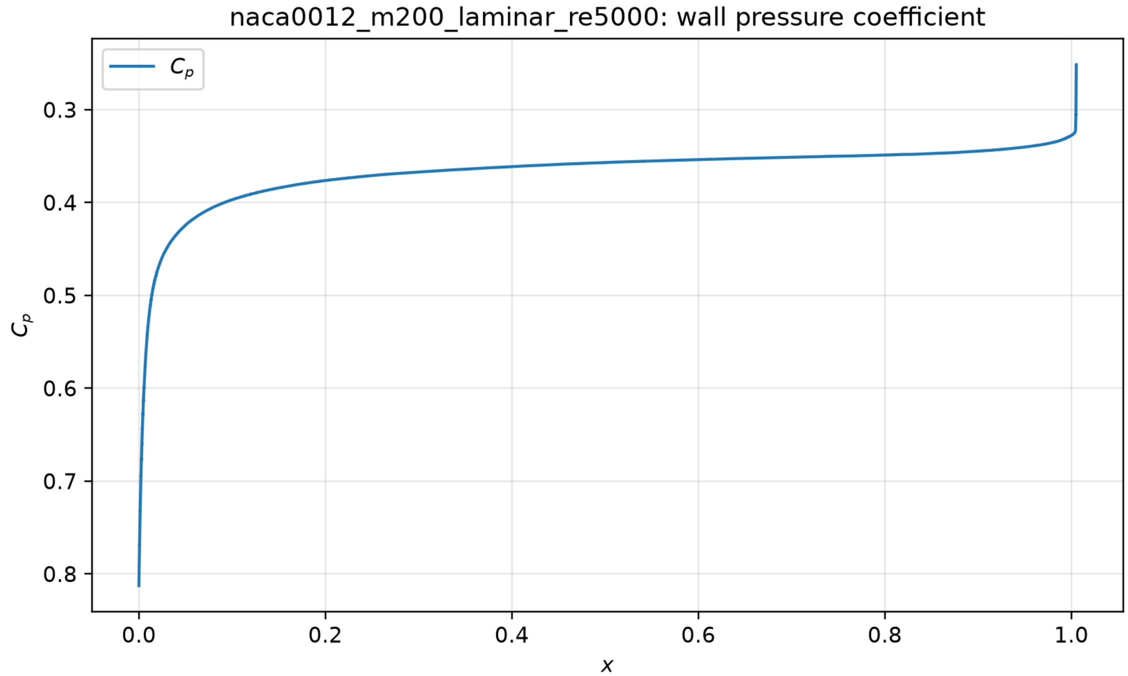

Figure 40: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

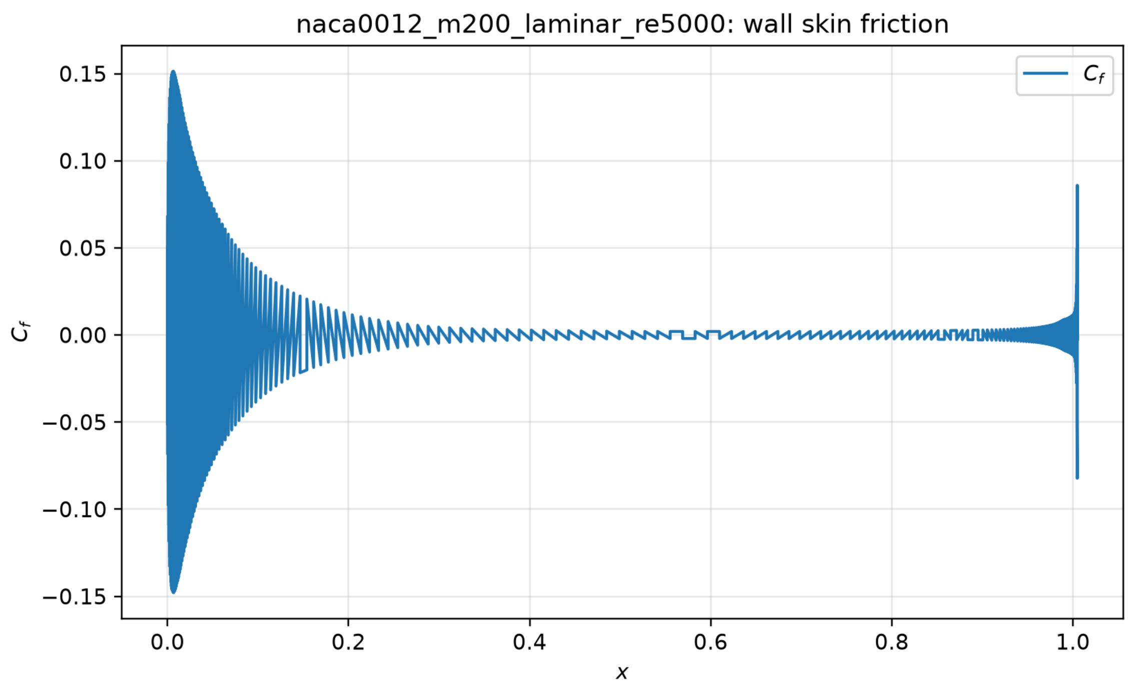

Figure 41: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

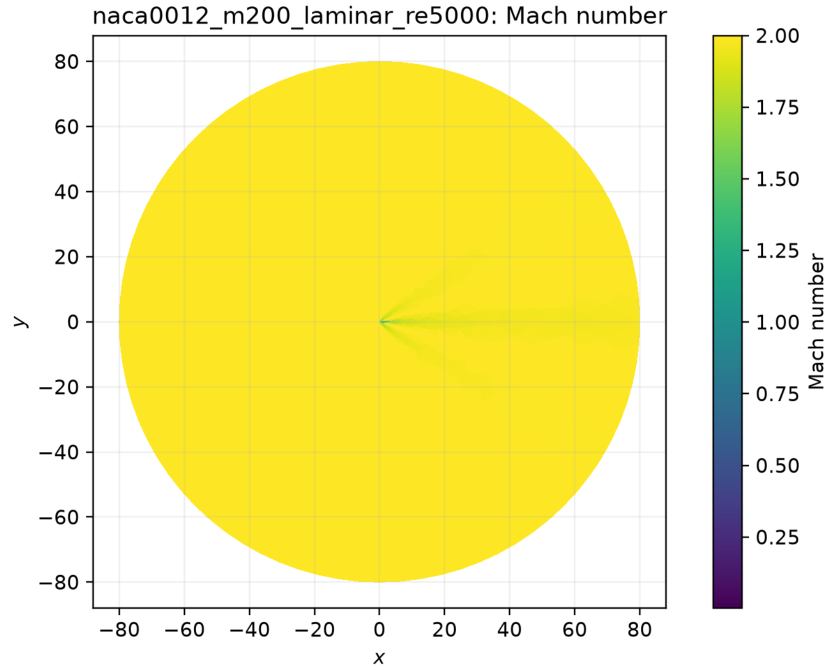

Figure 42: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

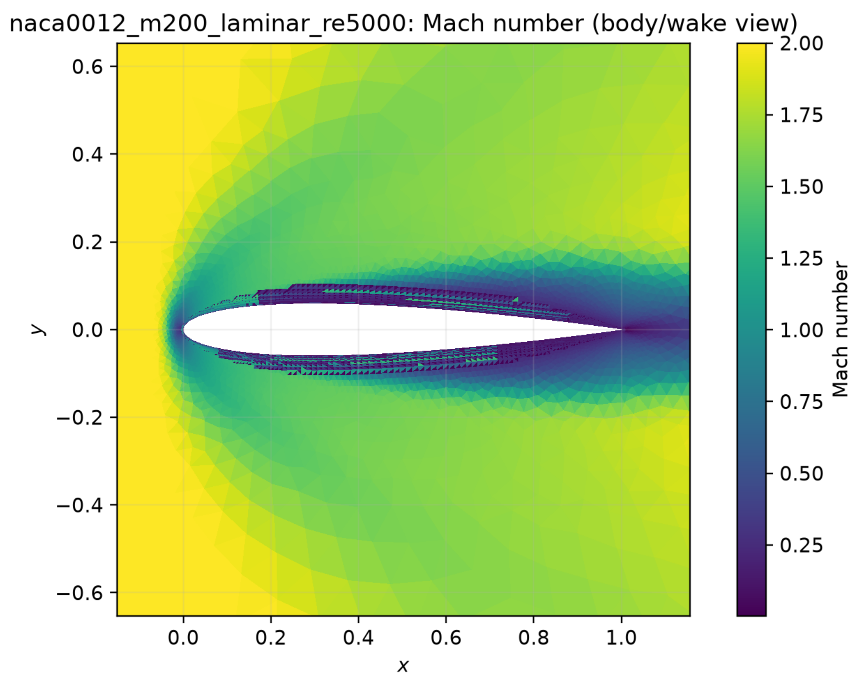

Figure 43: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

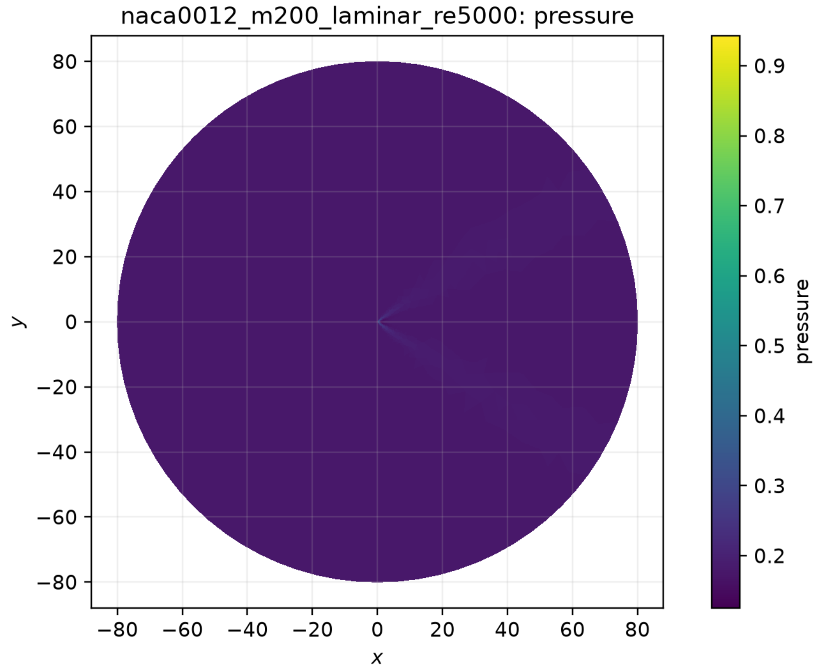

Figure 44: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

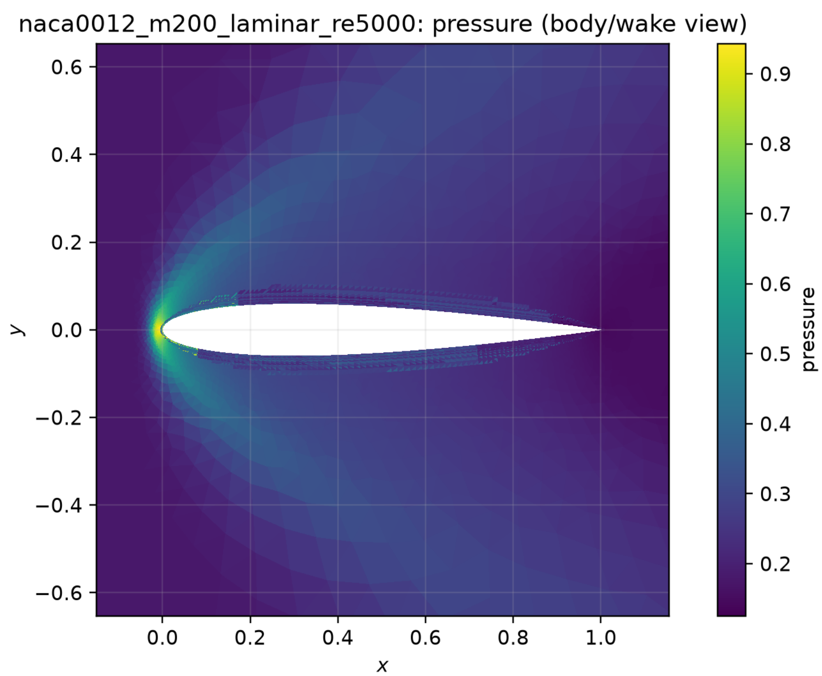

Figure 45: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

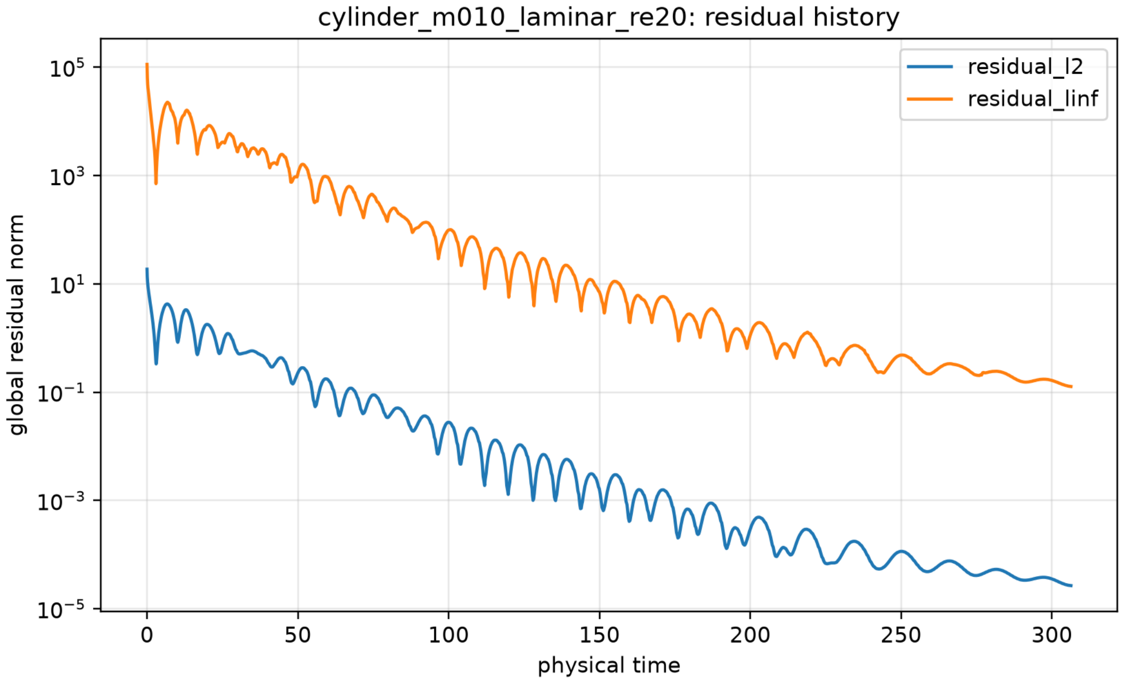

Figure 46: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

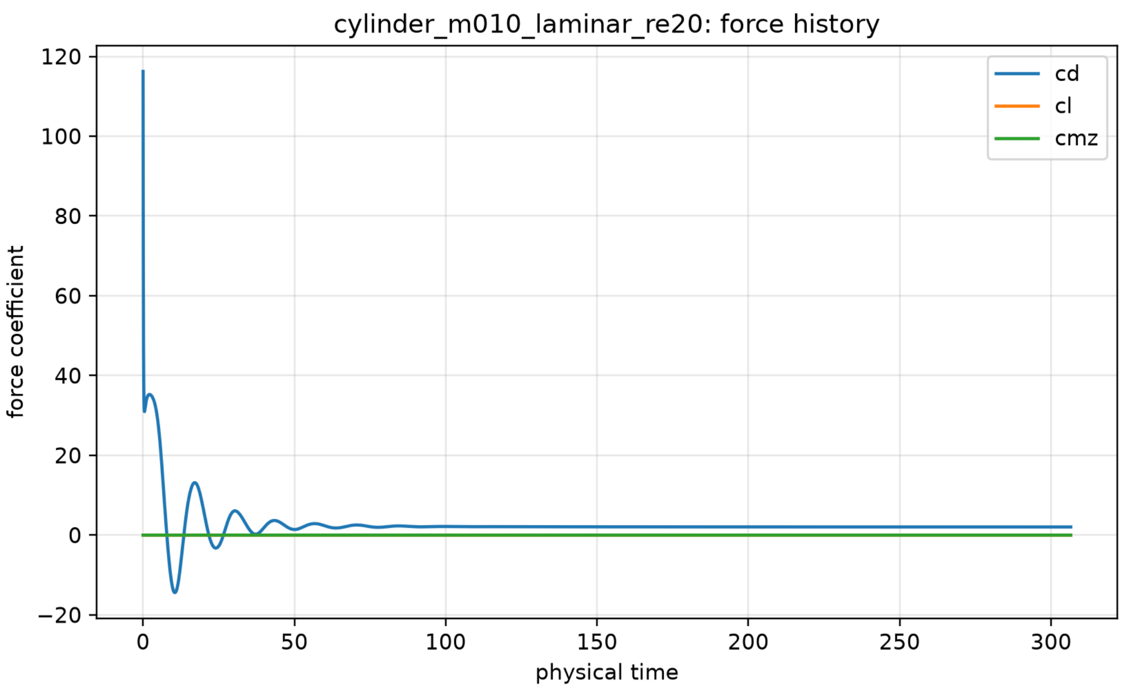

Figure 47: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

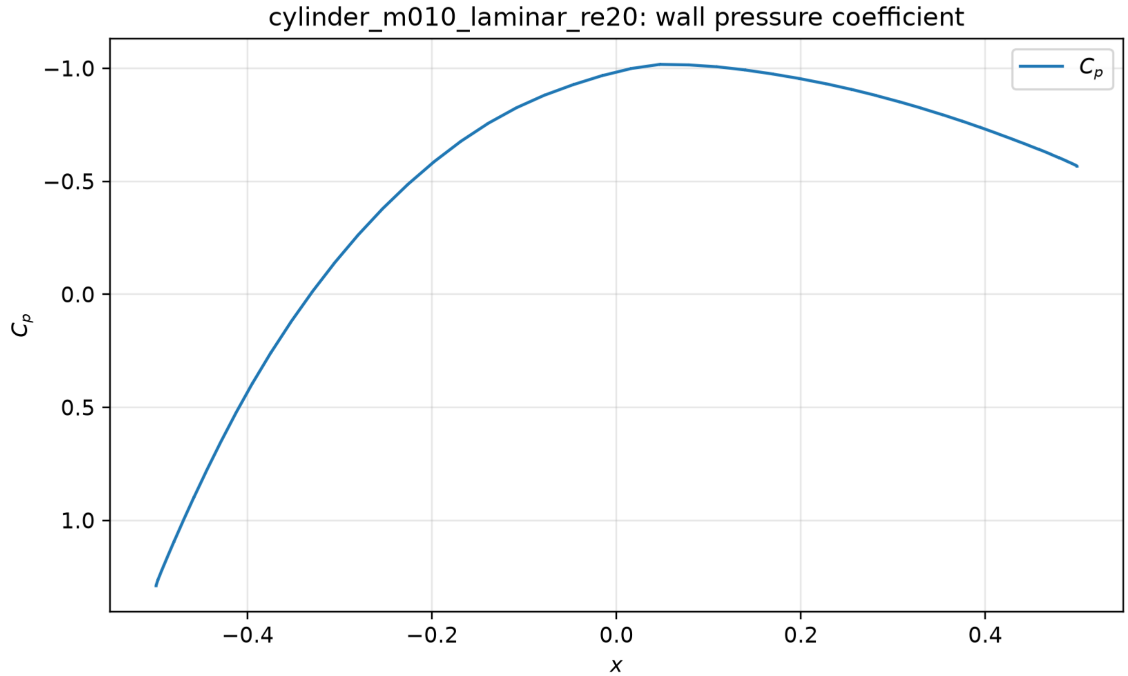

Figure 48: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

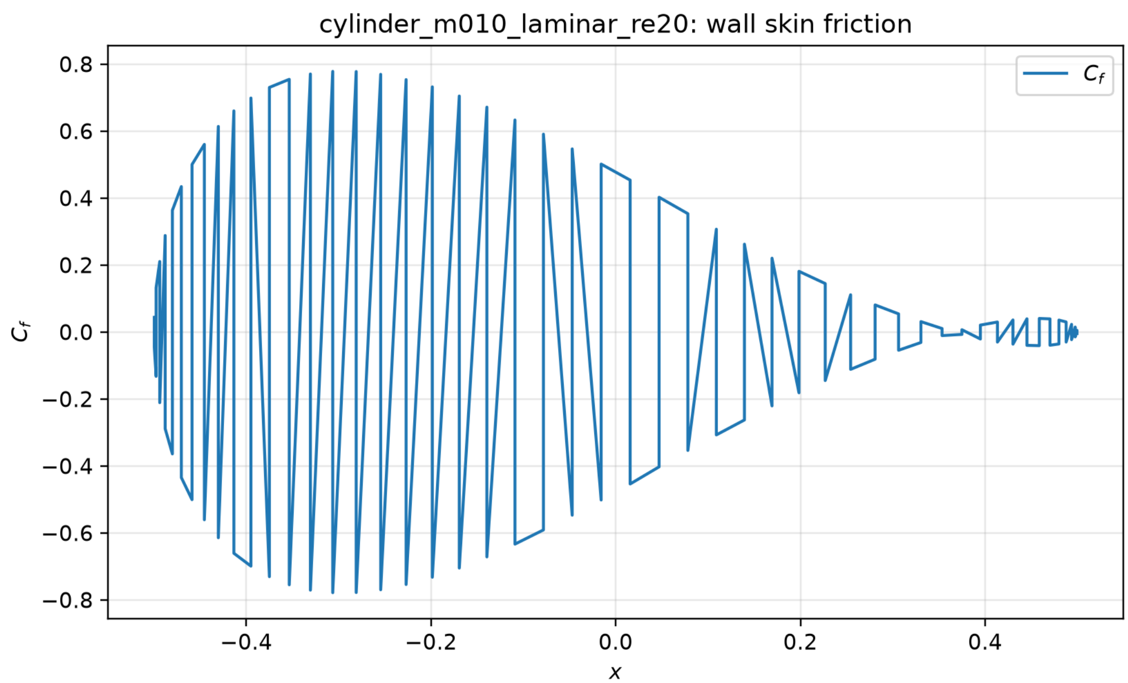

Figure 49: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

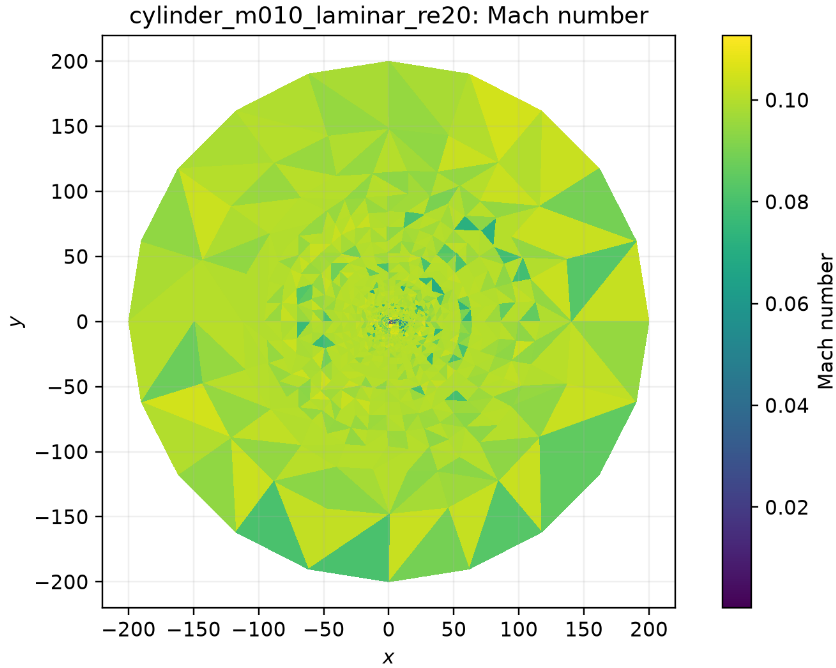

Figure 50: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

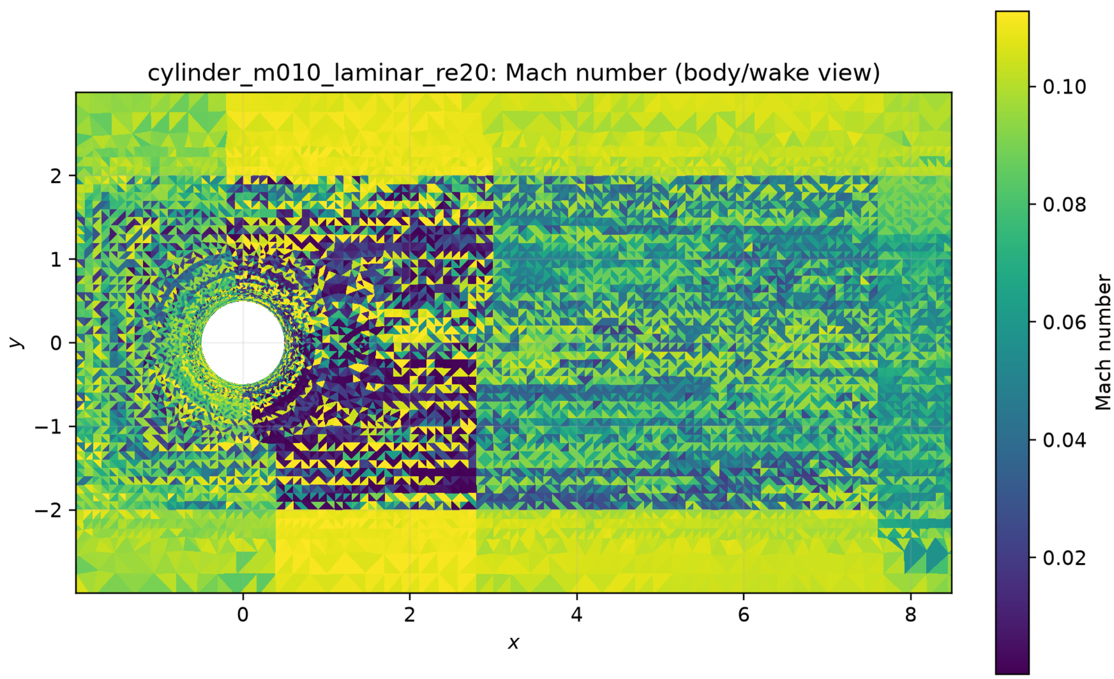

Figure 51: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

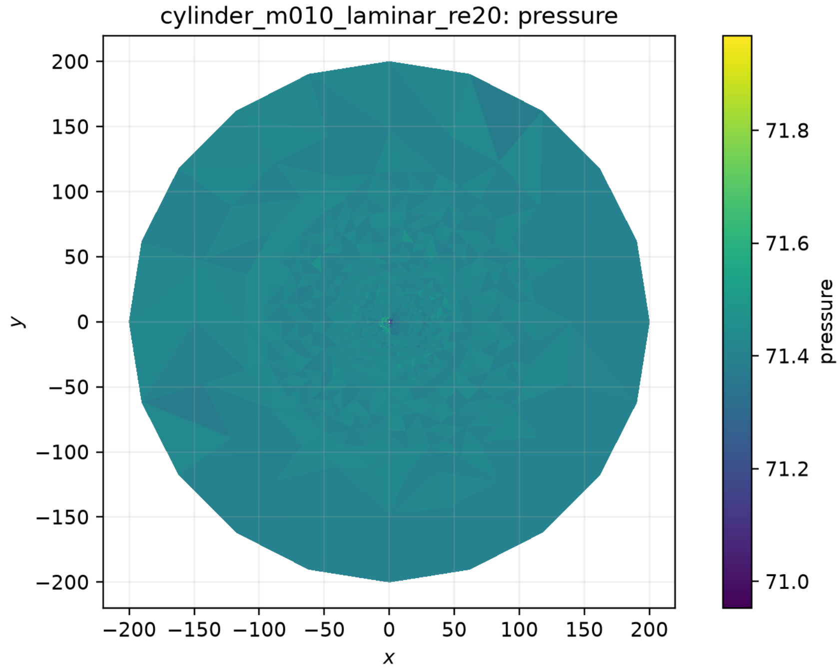

Figure 52: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

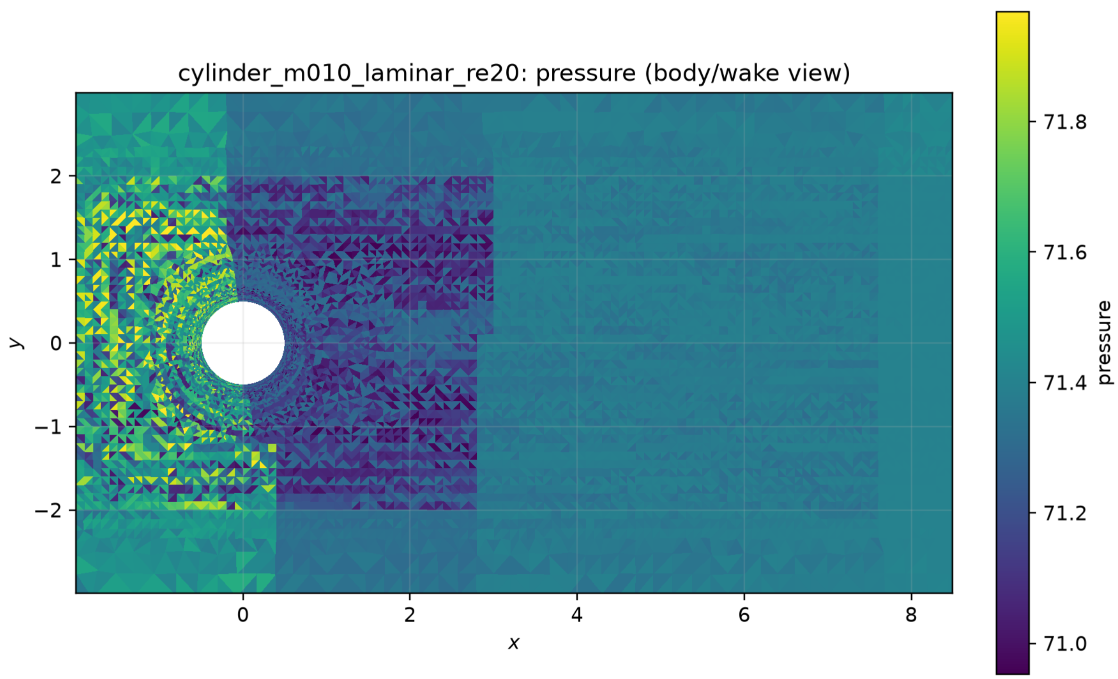

Figure 53: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

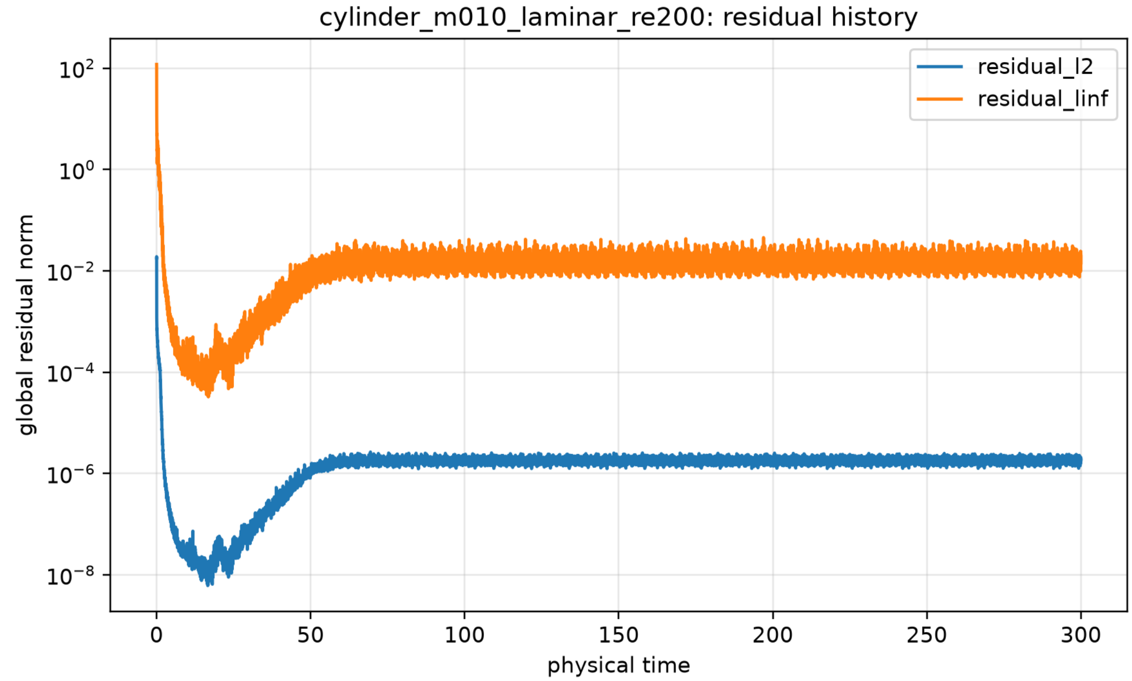

Figure 54: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

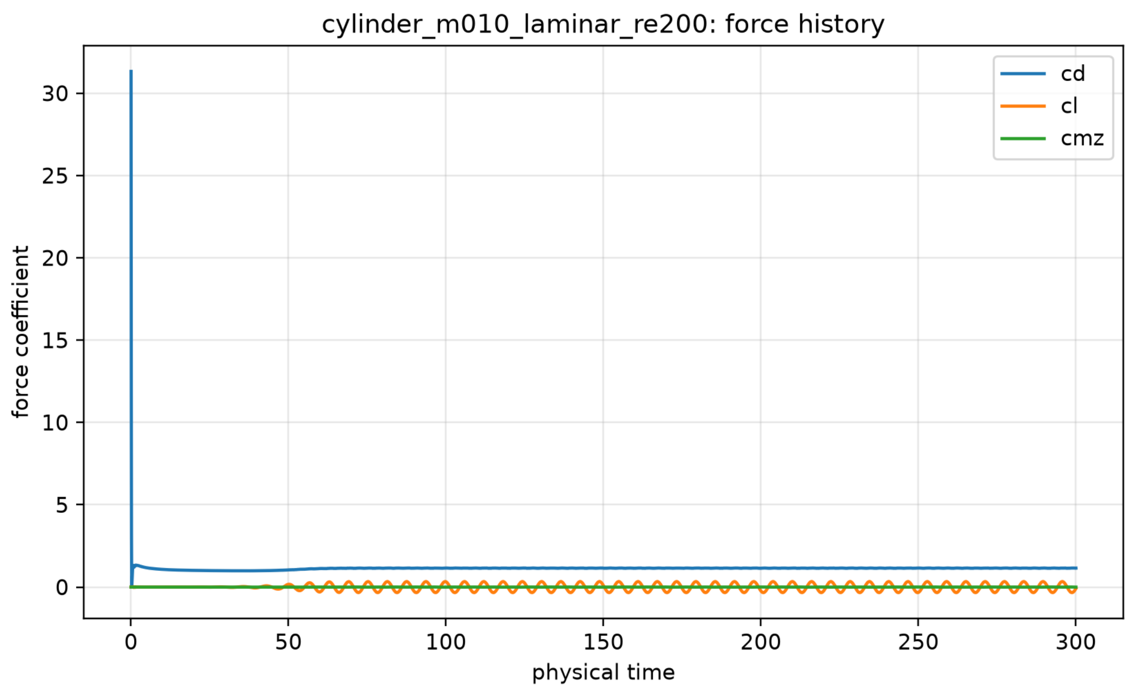

Figure 55: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

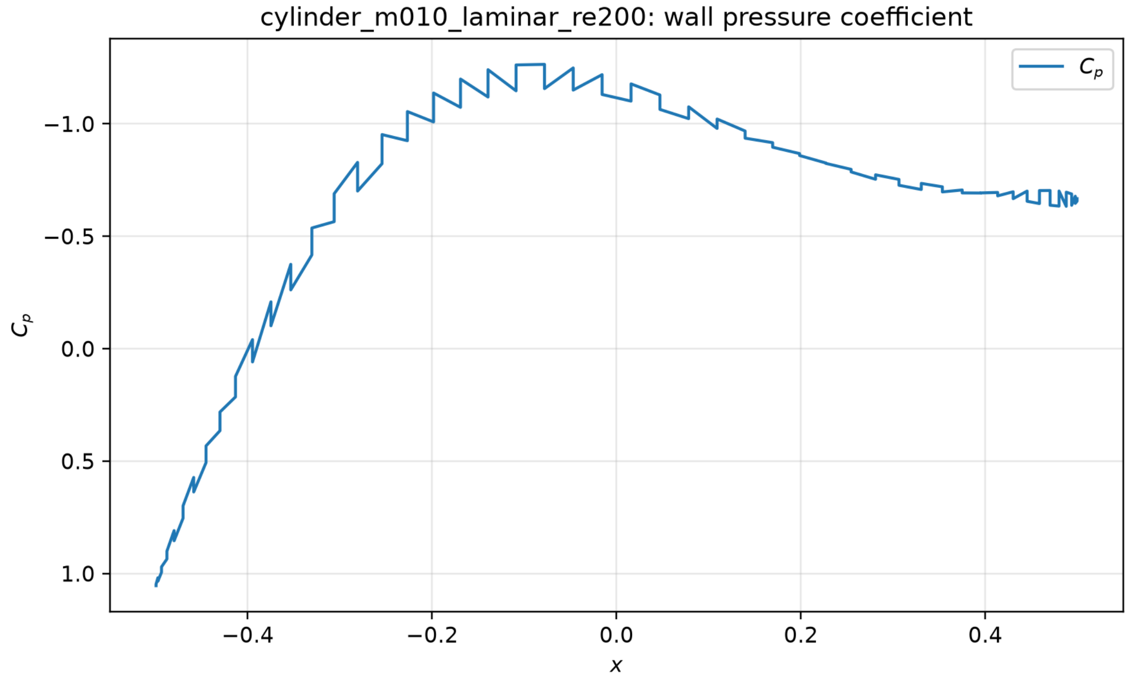

Figure 56: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

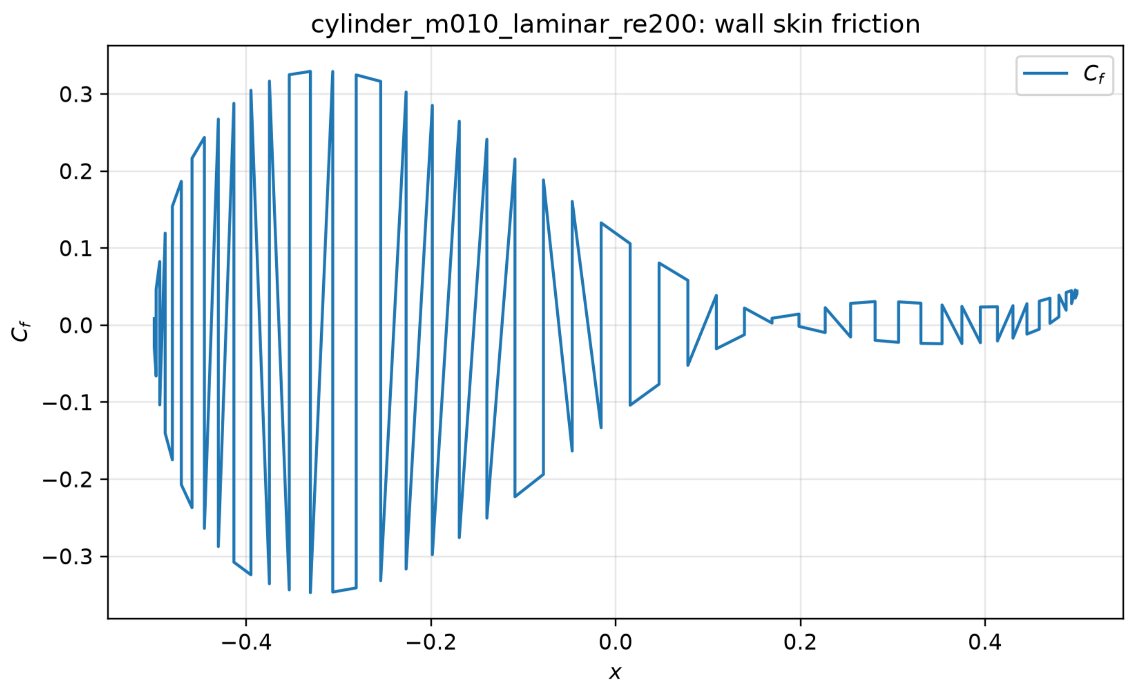

Figure 57: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

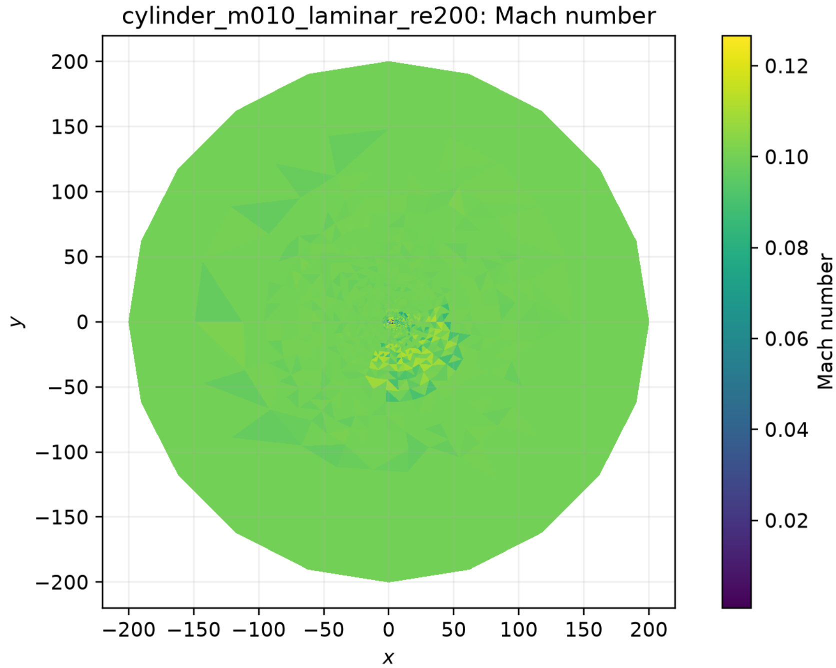

Figure 58: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

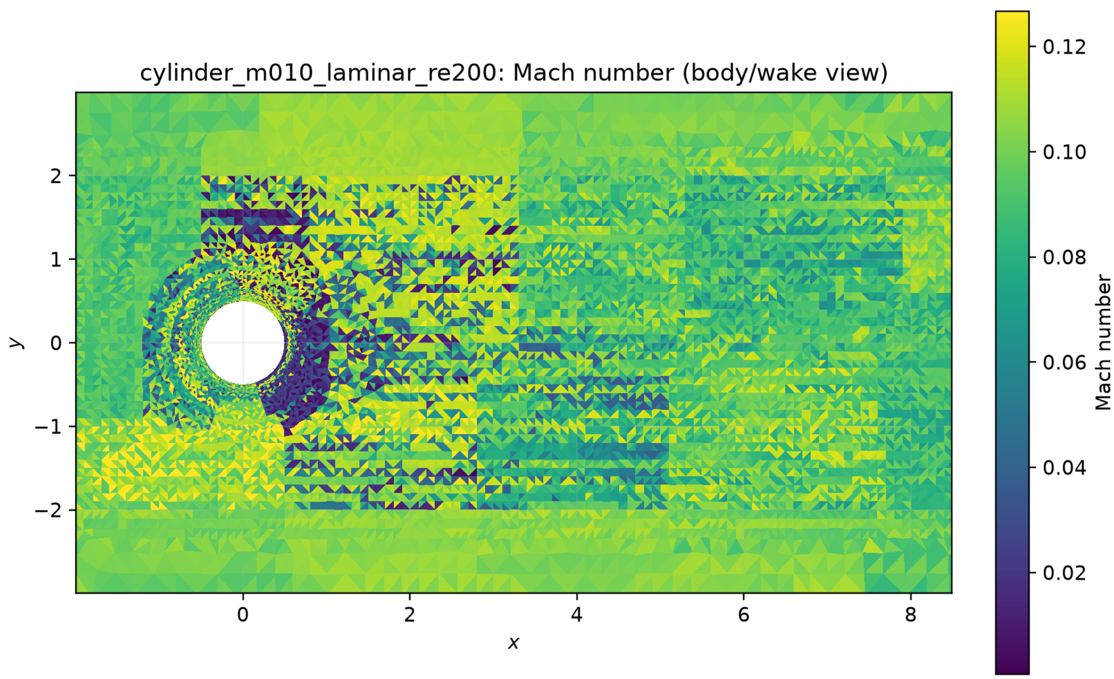

Figure 59: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

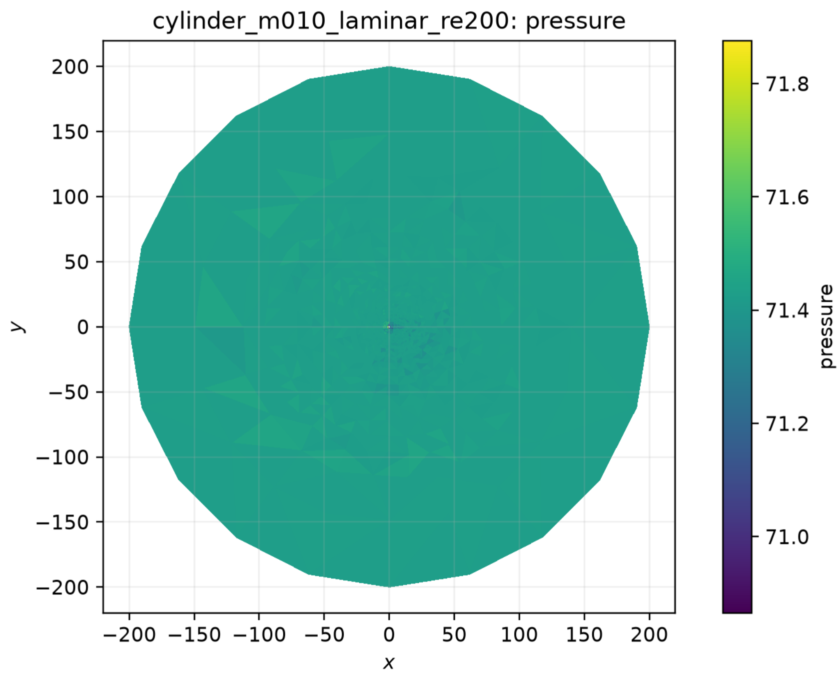

Figure 60: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

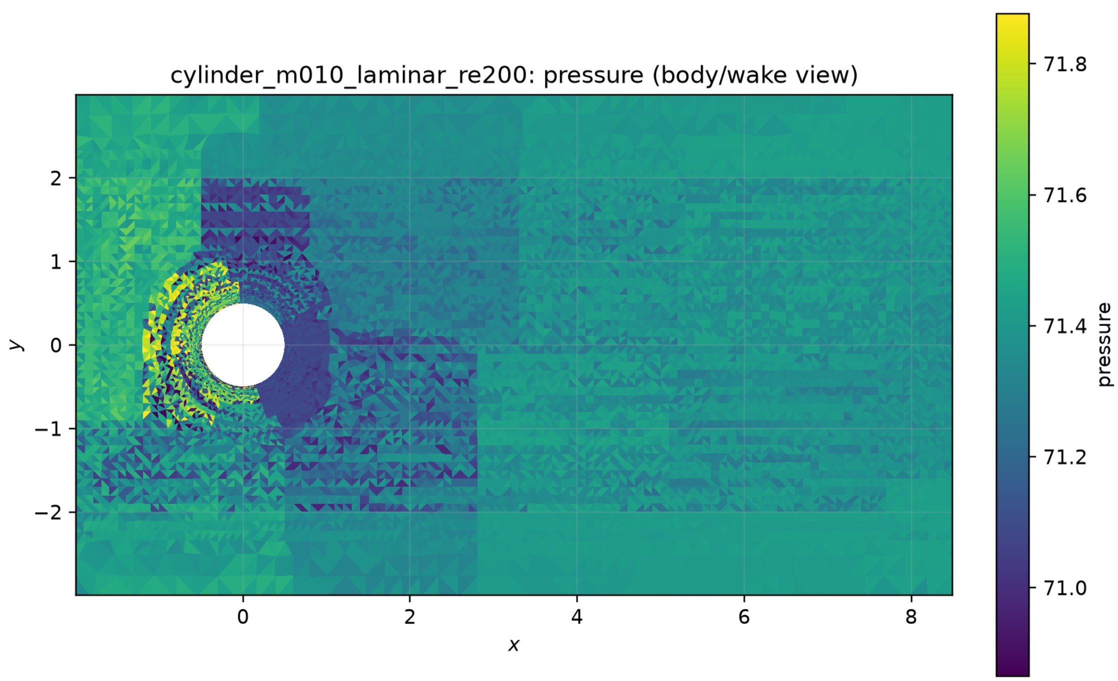

Figure 61: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

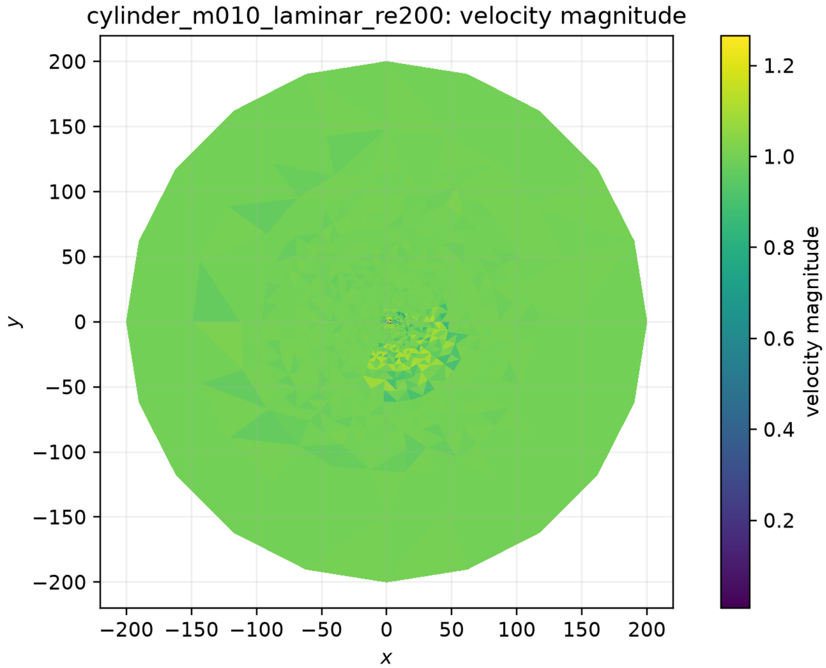

Figure 62: Post-transient wake visualization. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

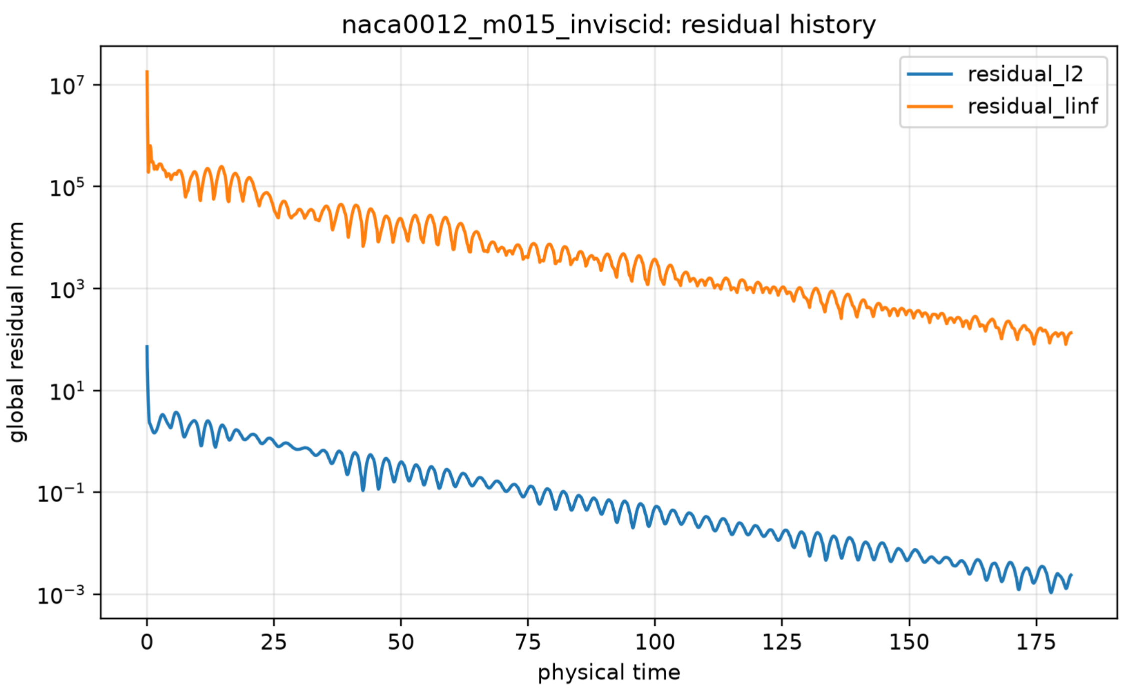

Figure 63: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

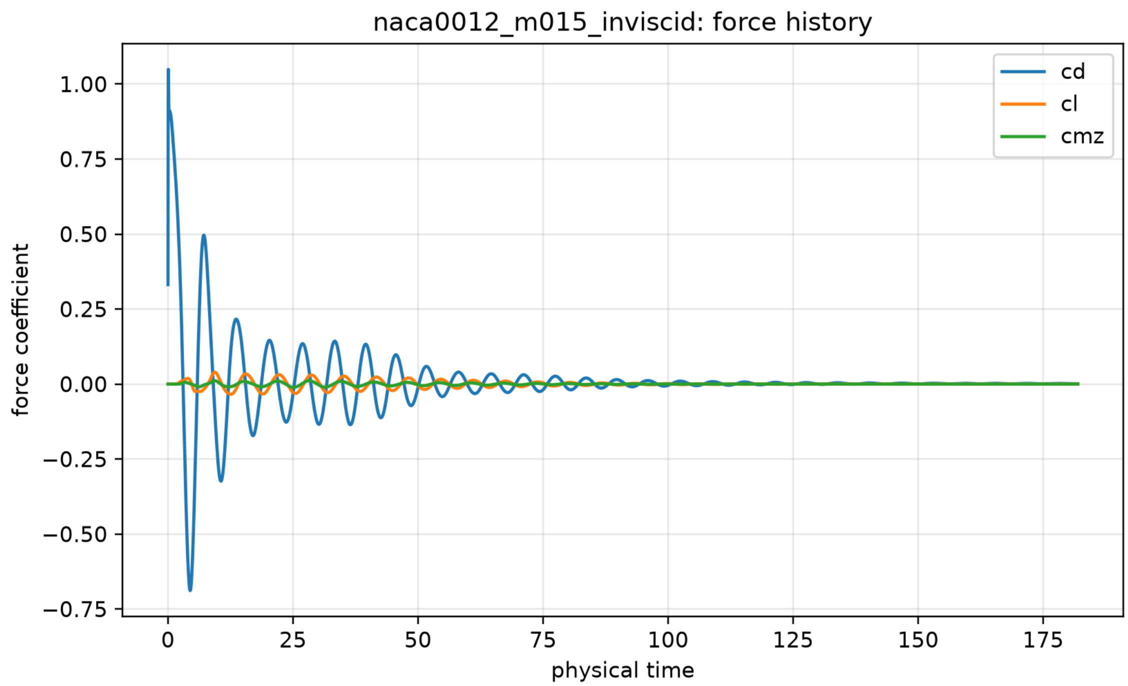

Figure 64: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

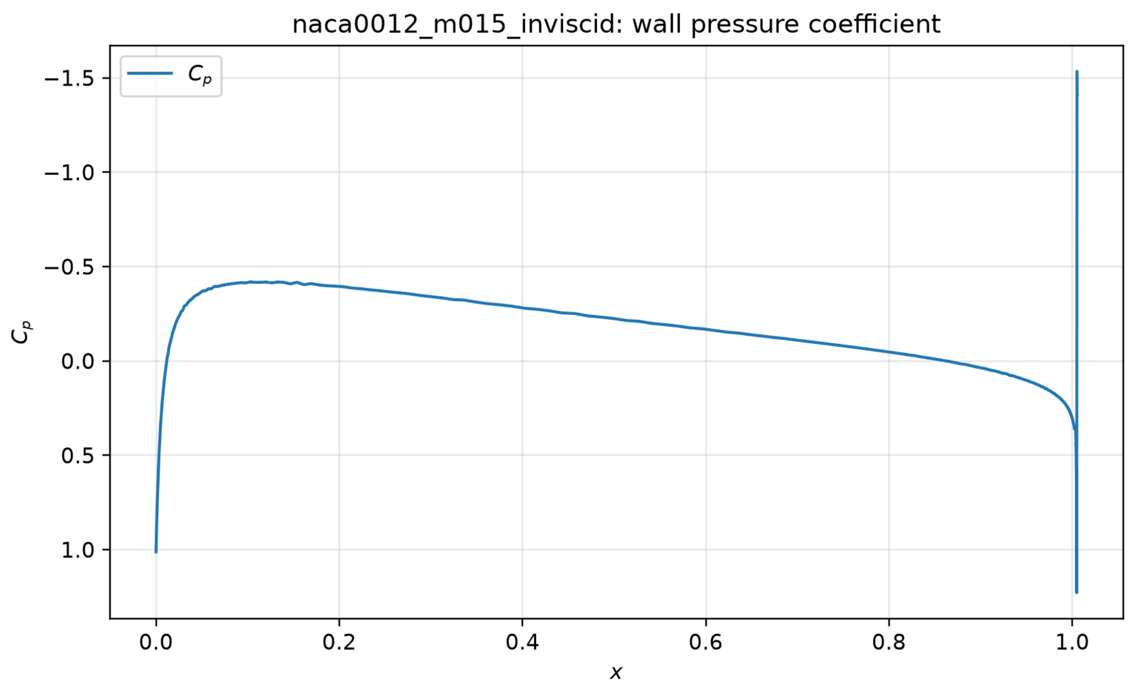

Figure 65: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

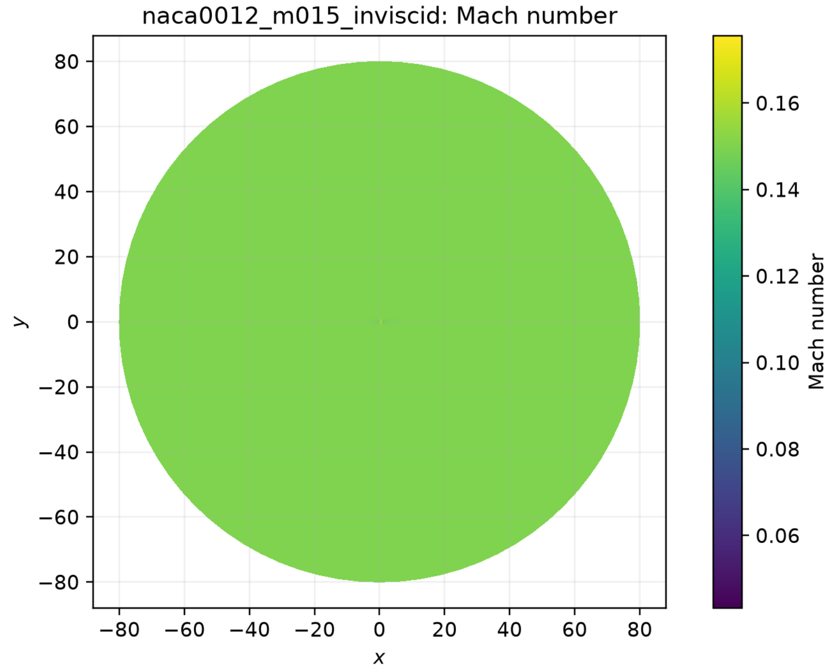

Figure 66: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

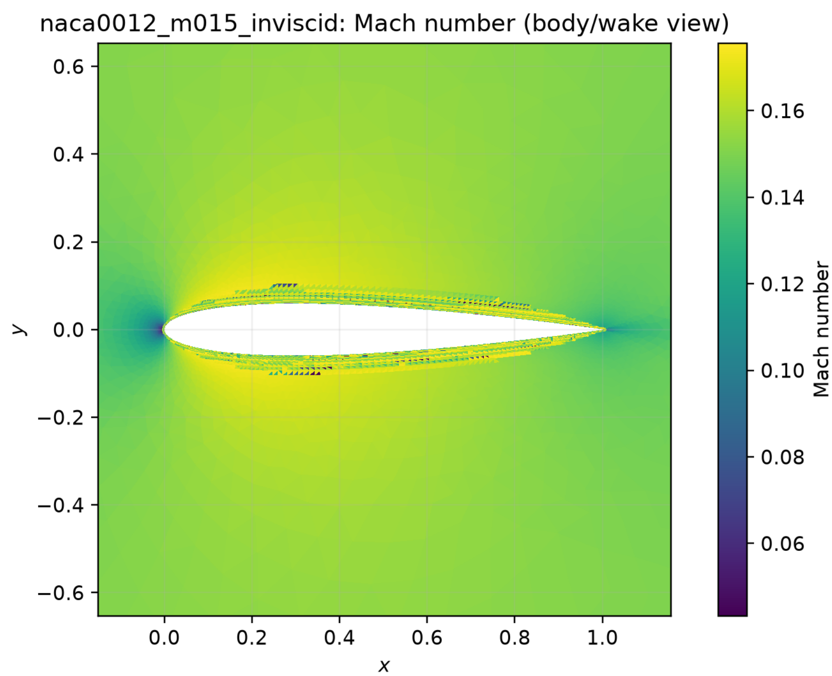

Figure 67: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

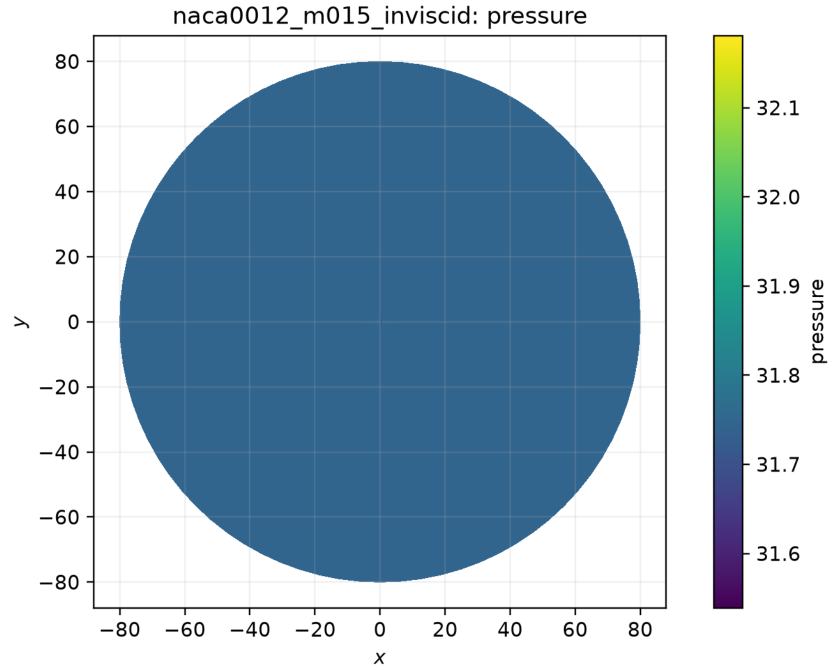

Figure 68: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

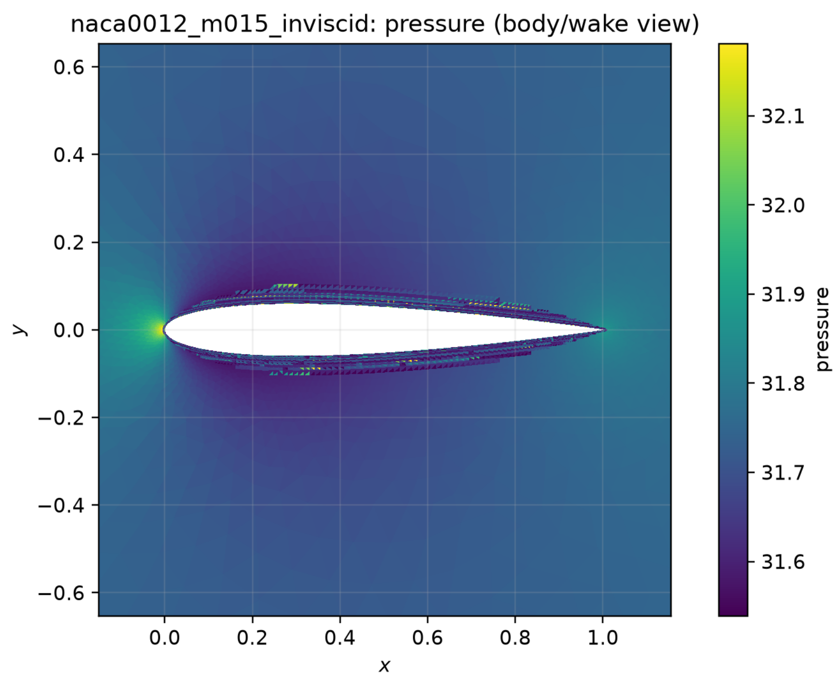

Figure 69: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

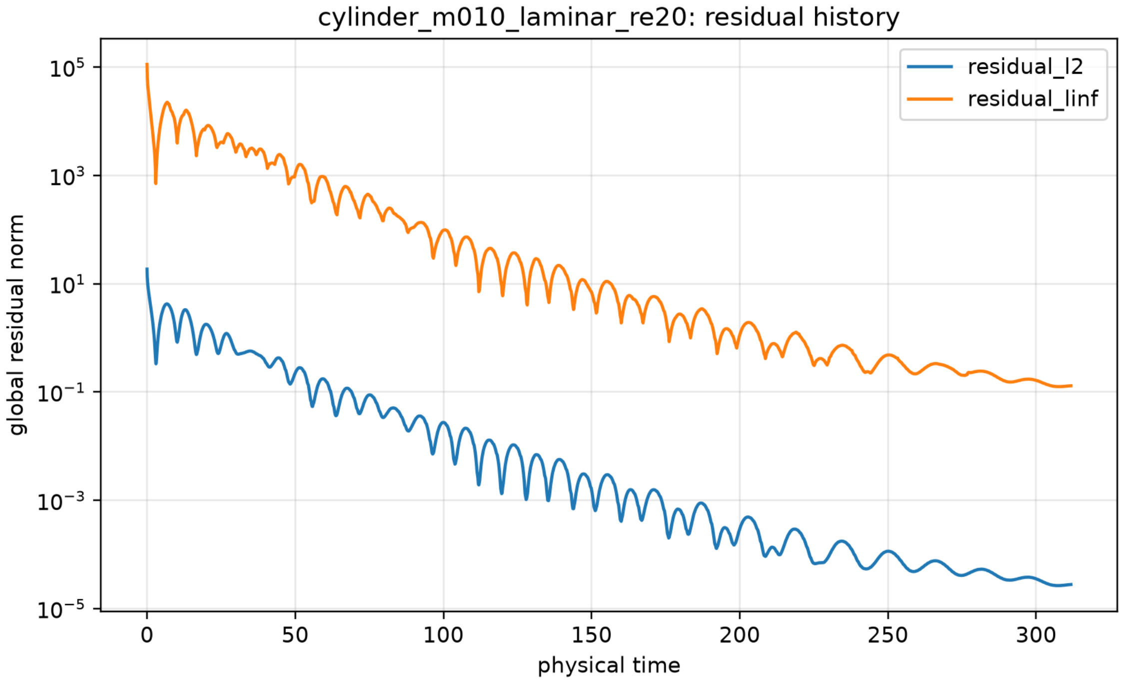

Figure 70: Global residual history. Source: `residuals.csv`; provenance: `figure_manifest.csv`.

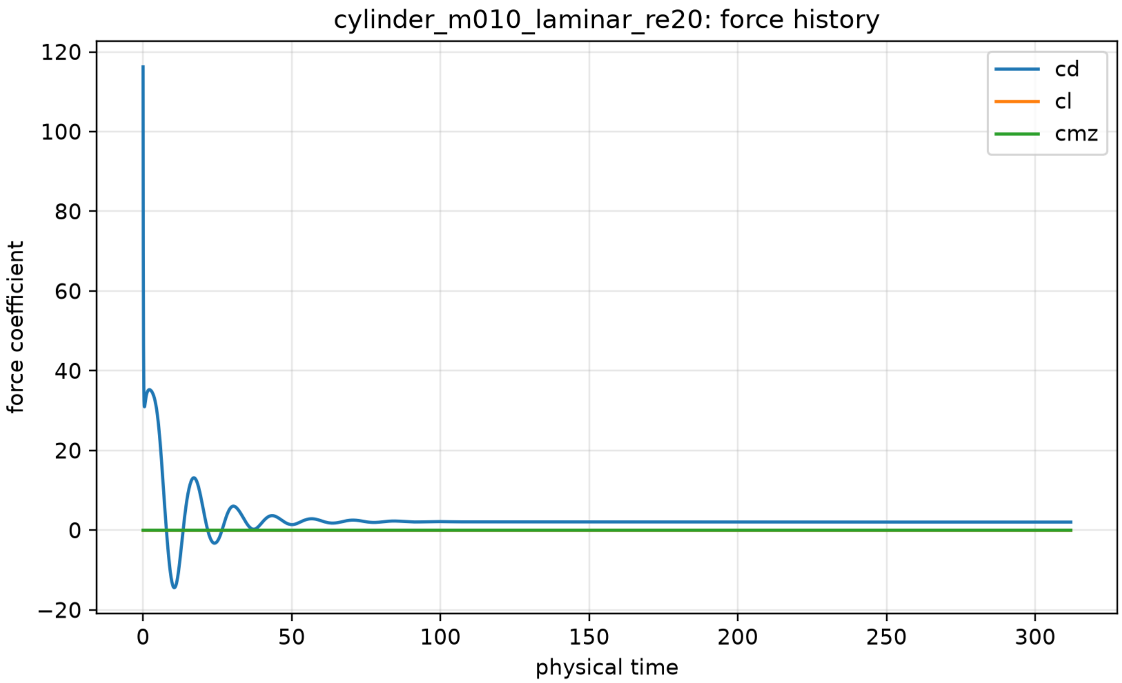

Figure 71: Force-coefficient history. Source: `forces.csv`; provenance: `figure_manifest.csv`.

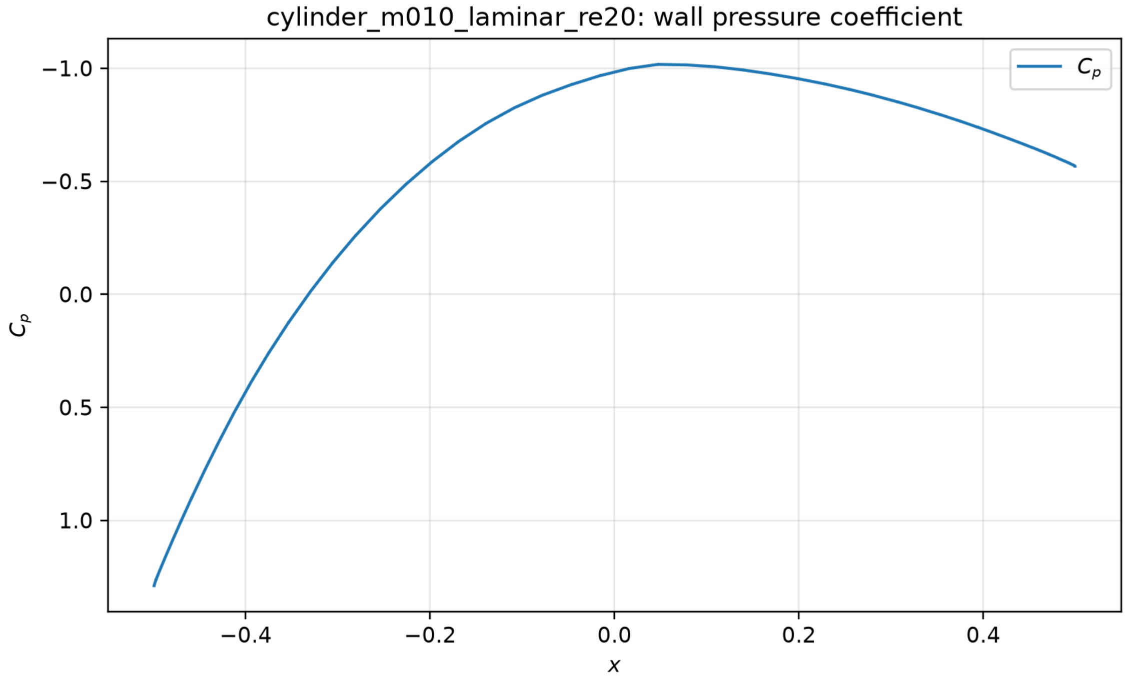

Figure 72: Wall pressure coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

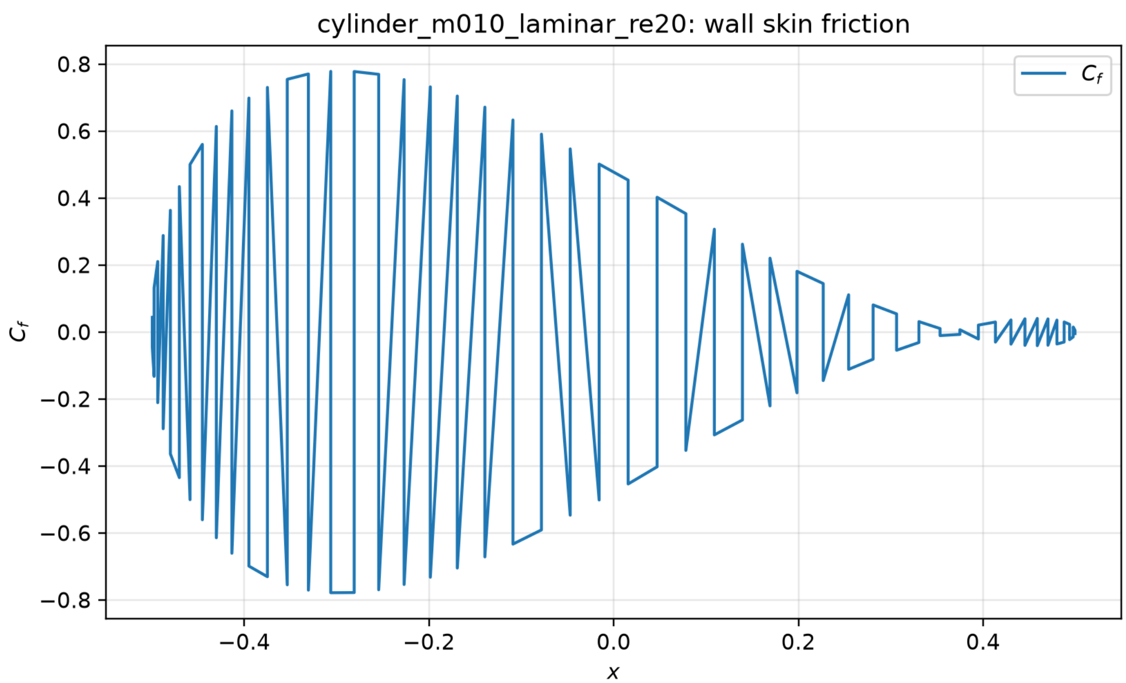

Figure 73: Wall skin-friction coefficient. Source: `surface.csv`; provenance: `figure_manifest.csv`.

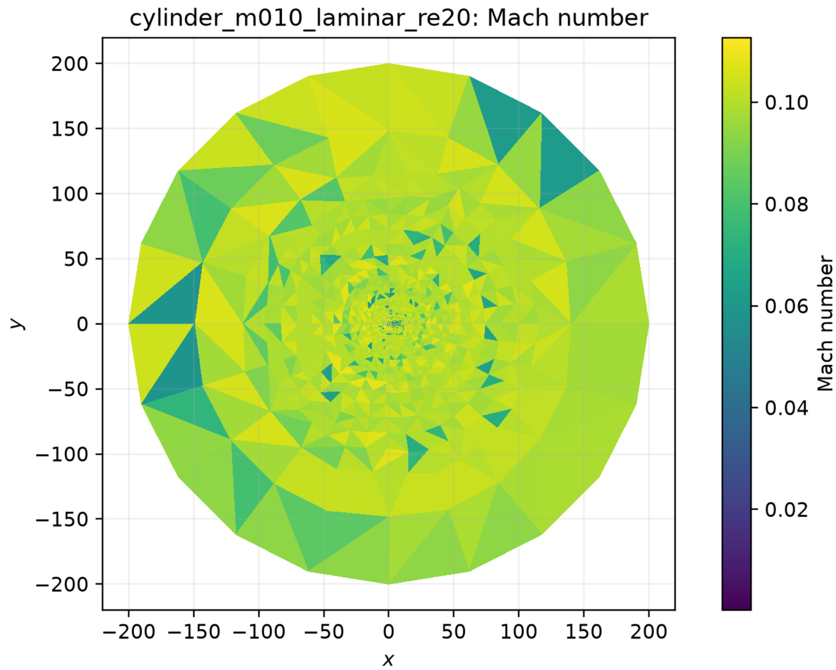

Figure 74: Filled Mach-number field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

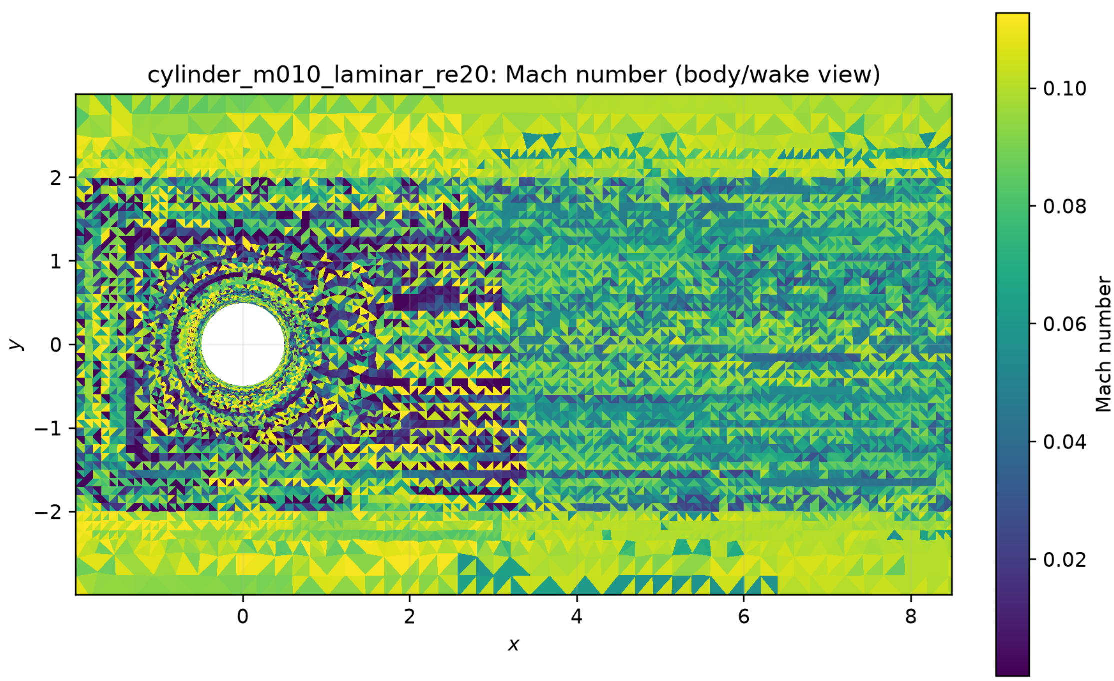

Figure 75: Filled Mach-number body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

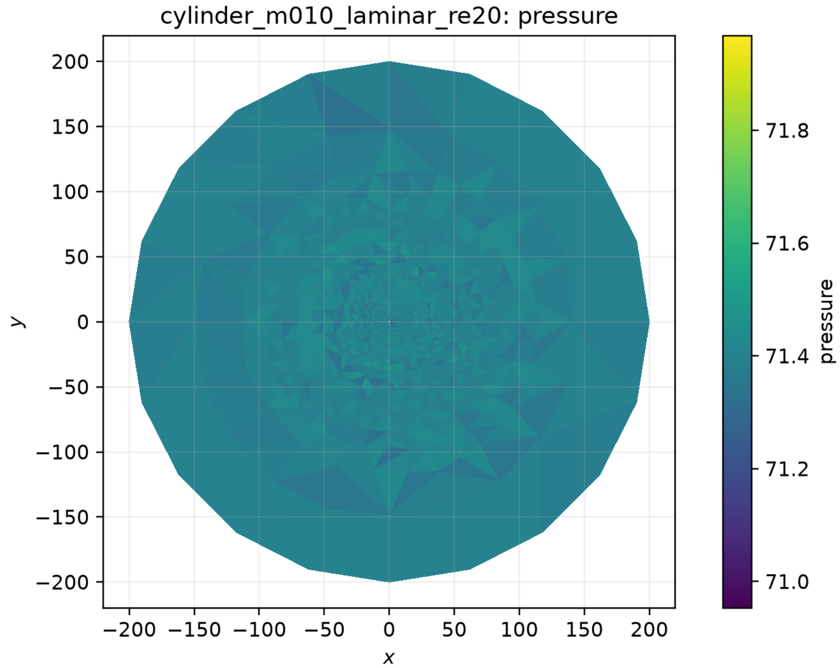

Figure 76: Filled pressure field. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

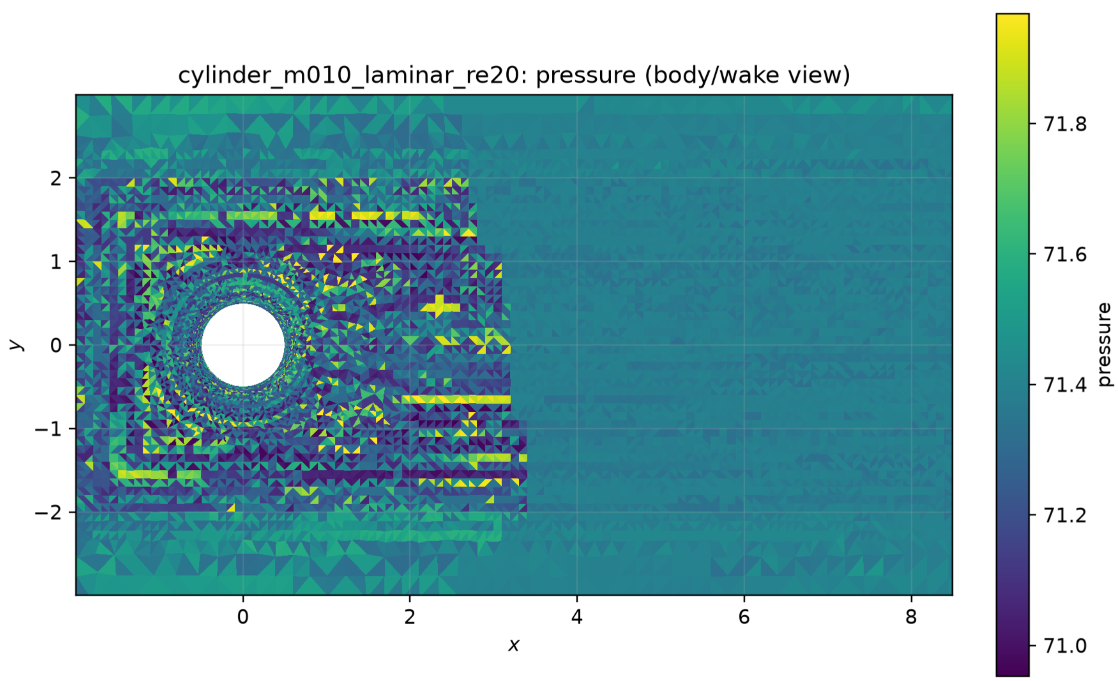

Figure 77: Filled pressure body or wake view. Source: `field_final.pvtu`; provenance: `figure_manifest.csv`.

7 Limitations

Any row qualified as a bounded-residual plateau is explicitly not evidence that the requested steady residual target was attained. Reported MPI comparisons are limited to actual paired runs and do not claim ideal scaling. The report makes no mesh-refinement, experimental-reference, turbulence-model, or formal discretization-error claim. In strong-compression cells, the pressure-ratio shock sensor also limits the viscous-gradient path; an inviscid-only-sensor ablation has not been submitted as part of this result set. The machine-readable sanity checks are a necessary positivity, wall-condition, and nontrivial-flow gate, not a substitute for independent physical validation.